Electromagnetic Theory

Lecture 1

20 August, 2024

1. Transmission Line
2. Vector Calculus
3. Static Electric field / Static Magnetic field
4. Maxwell's equation

Gauss's Law

Electric flux – $\phi = \frac{Q}{\varepsilon_0}$ – total Charge enclosed within volume (C)

Nm^2/C

ε_0 – Electric constant 8.85×10^{-12}

Integral: $\phi = \oint E \cdot ds$

Ampere Law $\mu_0 I$

Integral $\oint B\, dl = \mu_0 I$

differential $\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$

Faraday Law

differential

$$\nabla \times E = -\frac{d\vec{B}}{dt}$$

Integral

$$\oint_C \vec{E}\, dl = -\frac{d}{dt} \iint_S \vec{B} \cdot \hat{n}\, da$$

Gauss Law for magnetism

$$d\phi = B\, ds$$

$$\phi = \oint \vec{B}\, d\vec{s}$$

Gauss Law: The net flux through any closed surface equals the net charge inside that surface divided by ε_0

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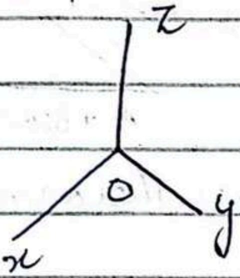
Lecture 2

Scalar	Vector
1. only magnitude	1. Magnitude and direction
2. one dimensional	2. multidimensional
3. Scalar can be divided by another scalar	3. Vector cannot be divided by another vector
Example Time, mass Length	Example velocity, weight, force

* vector cannot be divided by a vector $\frac{2ax + 3ay}{2ax + 3ay}$ (not possible)

* vector 3 dimensional can be rotated they are at angle of 90° from each other

* unit vector is represent by $\hat{i}_A$ always



Basics of Vector

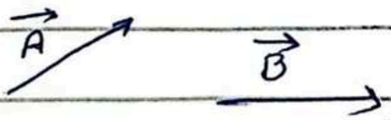
$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$$

$\vec{P} = (2, 3, 4)$
$\vec{Q} (3, 4, 5)$

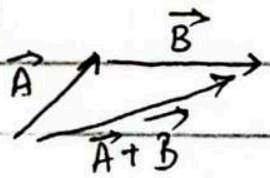
$$\vec{PQ} = Q - P = \sqrt{(3-2)^2 + (4-3)^2 + (5-4)^2}$$
$$= \sqrt{3}$$

$$|A| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

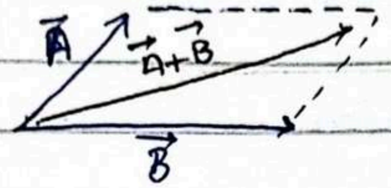
$$\text{Unit Vector} = \frac{\vec{A}}{|A|} = \frac{A_x\hat{i} + A_y\hat{j} + A_z\hat{k}}{\sqrt{A_x^2 + A_y^2 + A_z^2}} = \hat{i}_A$$



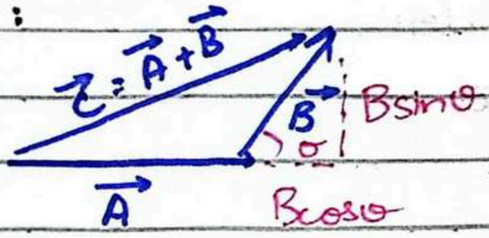
by head to tail rule

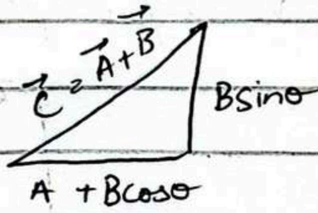


Parallelogram Law



Example:





by using pythogras theroam

$c^2 = a^2 + b^2$

$$c^2 = (A + B\cos\theta)^2 + (B\sin\theta)^2$$

$$C^2 = A^2 + B^2\cos^2\theta + 2AB\cos\theta + B^2\sin^2\theta$$

$$C^2 = A^2 + 2AB\cos\theta + B^2\cos^2\theta + B^2\sin^2\theta$$

$$C^2 = A^2 + 2AB\cos\theta + B^2(\cos^2\theta + \sin^2\theta)$$

$$C^2 = A^2 + 2AB\cos\theta \quad B^2(1)$$

$$\boxed{C^2 = A^2 + B^2 + 2A\cos\theta}$$

Law of parallelogram

Dot product:

$$\vec{C} = \vec{A} \cdot \vec{B}$$

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$$

$$\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$$

$\vec{A}\cdot\vec{B} = |A|\,|B|\cos\theta$

$\vec{A}\cdot\vec{B} = A_xB_x + A_yB_y + A_zB_z$

θ is the angle between both $\vec{A}$ and $\vec{B}$

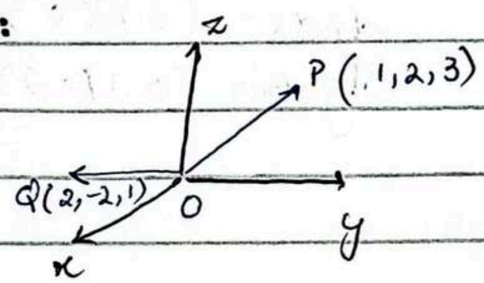
At a point there is intersection of planes each plane intersect each other at a point

Cross Product:

$\vec{Y} = \vec{A}\times\vec{B} = |A|\,|B|\sin\theta$

$$\vec{Y} = \vec{A}\times\vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

Example:



$\vec{P} = 1\hat{i} + 2\hat{j} + 3\hat{k}$

$\vec{Q} = 2\hat{i} - 2\hat{j} + 1\hat{k}$

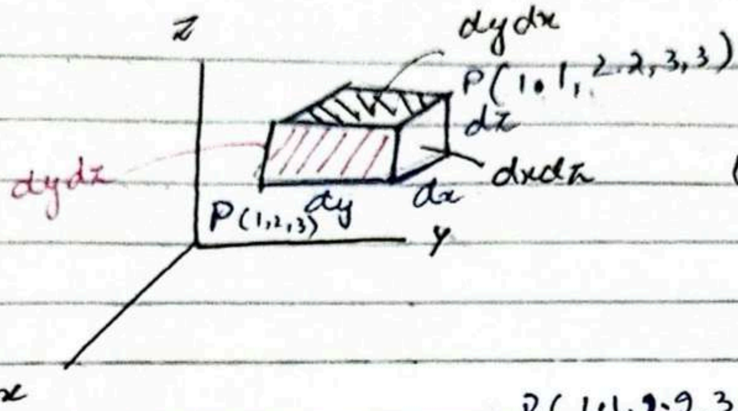
$\vec{r}_{PQ} = (2\hat{i} - 2\hat{j} + 1\hat{k}) - (1\hat{i} + 2\hat{j} + 3\hat{k})$

$\vec{r}_{PQ} = 1\hat{i} - 4\hat{j} - 2\hat{k}$

$|\vec{r}_{PQ}| = \sqrt{(1)^2 + (4)^2 + (2)^2}$

$|\vec{r}_{PQ}| = \sqrt{1 + 16 + 4}$

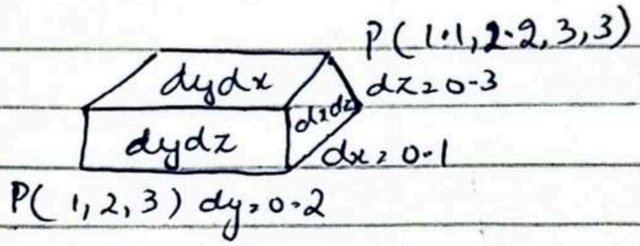
$|\vec{r}_{PQ}| = \sqrt{21}$



$dv = dx\,dy\,dz$

$|dV| = \sqrt{dx^2 + dy^2 + dz^2}$

Example:



$dv = dx\,dy\,dz$

$= 0.1 \times 0.2 \times 0.3$

$\boxed{dv = 0.006 \text{ units}}$

$dy\,dz = (0.2)(0.3) = 0.06$

$dx\,dz = (0.1)(0.3) = 0.03$

$dy\,dx = (0.1)(0.2) = 0.02$

$|dv| = \sqrt{0.1^2 + 0.2^2 + 0.3^2}$

$\boxed{|dv| = 0.374 \text{ units}}$

Example Specify the unit vector from the origin towards the point $F(2, -2, 1)$

$\vec{OF} = F - O = (2,-2,1) - (0,0,0)$

$\vec{OF} = (+2, -2, +1)$

$|\vec{OF}| = \sqrt{2^2 + 2^2 + 1} = \sqrt{4+4+1} = 3$

$\hat{e}_F = \dfrac{\vec{FO}}{|\vec{FO}|} = \dfrac{2\hat{i} - 2\hat{j} + \hat{k}}{3} = \dfrac{2}{3}i - \dfrac{2}{3}k + \dfrac{1}{3}\hat{k}$

CamScanner

Example $U(-1,2,1)$, $L(3,-3,0)$, $S(-2,-3,-4)$

$\vec{R}_{UL} = ?$, $\vec{R}_{UL} + \vec{R}_{US}$, $|r_U| = ?$ $|r_S| = ?$

Unit vector of $R_{US} = ?$

$\vec{U} = -1\hat{i} + 2\hat{j} + 1\hat{k}$ $\vec{L} = 3\hat{i} - 3\hat{j}$ $S = -2\hat{i} - 3\hat{j} - 4\hat{k}$

$\vec{R}_{UL} = \vec{L} - \vec{U} = (3\hat{i} - 3\hat{j}) - (-1\hat{i} + 2\hat{j} + 1\hat{k})$

$= (3+1)\hat{i} + (-3-2)\hat{j} + (0-1)\hat{k}$

$= 4\hat{i} - 5\hat{j} - 1\hat{k}$

$\vec{R}_{US} = \vec{S} - \vec{U} = (-2\hat{i} - 3\hat{j} - 4) - (-1\hat{i} + 2\hat{j} + 1\hat{k})$

$= (-2+1)\hat{i} + (-3-2)\hat{j} + (-4-1)\hat{k}$

$\vec{R}_{US} = -\hat{i} - 5\hat{j} - 5\hat{k}$

$\vec{R}_{UL} + \vec{R}_{US} = (4i - 5j - 1k) + (-i - 5j - 5k)$

$\vec{R}_{UL} + \vec{R}_{US} = 3\hat{i} - 10\hat{j} - 6k$

$|r_U| = \sqrt{(1)^2 + (2)^2 + (1)^2} = \sqrt{1+4+1}$

$|r_U| = \sqrt{6}$

$|r_S| = \sqrt{2^2 + 3^2 + 4^2} = \sqrt{4+9+16} = \sqrt{29}$

$|r_S| = \sqrt{29}$

$|\vec{R}_{US}| = \sqrt{(1)^2 + (5)^2 + (5)^2} = \sqrt{1+25+25} = \sqrt{51}$

27 August 2024

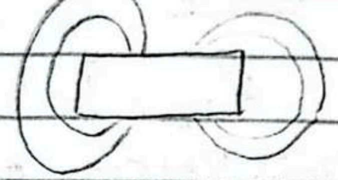
Lecture 3

Cylinderical Coordinate System

Real world Application

- Electromagnetics
- Fluid dynamics
- Electromagnetic field in wires
- Flow in pipes
- Heat conduction in cylinders

Cylinderical Coordinate System	Rectangular Coordinate System
• Define by ρ, ϕ and z	• Define by x, y and z axis
• ρ (radical distance from the z axis)	
• ϕ (Angle it makes in xy plane)	
• z (distance along z axis)	
• Surfaces of constant ρ are centered along z axis	
• Surface of constant ϕ are planes passing through z axis	
• surfaces of constant z are horizontal planes.	

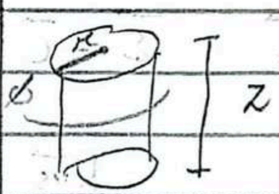


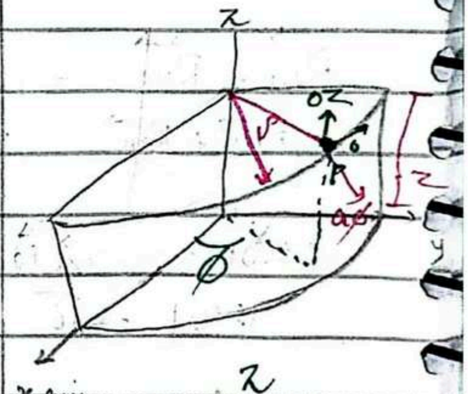
this electric waveform can be measure in rectangular coordinate system

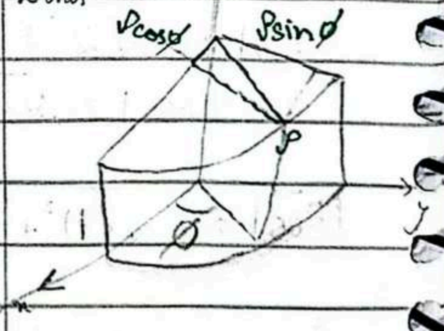
z axis is same in rectangular and cylinderical

ρ define r (radius) → always constant

ϕ define θ

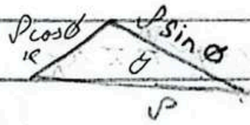






Conversion from RCS to CCS

$\rho = \sqrt{x^2+y^2}$



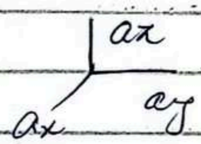
$\tan\phi = \frac{y}{x}$

$\phi = \tan^{-1}\left(\frac{y}{x}\right)$

$z = z$

RCS
↳ Rectangular coordinate system

CCS
↳ cylindrical coordinate system



conversion from CCS to RCS

ρ, ϕ and z are given

So
$x = \rho\cos\phi$
$y = \rho\sin\phi$
$z = z$

Unit Vectors

- a_ρ , a_ϕ , a_z (CCS)
- a_x , a_y , a_z (RCS)

Transforming Unit Vectors

	a_ρ	a_ϕ	a_z
a_x	$\cos\phi$	$-\sin\phi$	0
a_y	$\sin\phi$	$\cos\phi$	0
a_z	0	0	1

$A = A_\rho a_\rho + A_\phi a_\phi + A_z a_z$

(expression of RCS)

$A = A_x a_x + A_y a_y + A_z a_y$

(expression of CCS)

$A_{rect} = A_x a_x + A_y a_y + A_z a_z$ this is given we want to convert to cylindrical

$A_{cy} = A_\rho a_\rho + A_\phi a_\phi + A_z a_z$

$A_\rho = ?$ $A_\phi = ?$ $A_z = ?$

To find any desire component of the vector, we take the dot product of the vector and a unit in desire direction a_ρ

$A_\rho = A_{rect} \cdot a_\rho$

$A_\phi = A_{rect} \cdot a_\phi$

$A_z = A_z$

$A_\rho = (A_x a_x + A_y a_y + A_z a_y) \cdot a_\rho$
$= A_x (a_x a_\rho) + A_y (a_y a_\rho) + A_z (a_y a_\rho)$
$= A_x \cos\phi + A_y \sin\phi + A_z (0)$

$\boxed{A_\rho = A_x \cos\phi + A_y \sin\phi}$

$A_\phi = A_{rect} \cdot a_\phi$

$= (A_x a_x + A_y a_y + A_z a_z) \cdot a_\phi$

$= A_x (a_x \cdot a_\phi) + A_y (a_y \cdot a_\phi) + A_z (a_z \cdot a_\phi)$

$= A_x (-\sin\phi) + A_y (\cos\phi) + A_z (0)$

$\boxed{A_\phi = -A_x \sin\phi + A_y \cos\phi}$

perpendicular ka dot product is equal to zero

$\boxed{A_z = A_z}$

$B = y a_x - x a_y + z a_z$

transform it into cylindrical coordinates

$B_{cyl} = B_\rho a_\rho + B_\phi a_\phi + B_z a_z$

$B_\rho = ? \quad B_\phi = ? \quad B_z = ?$

$B_\rho = B \cdot a_\rho$

$B_\phi = B \cdot a_\phi$

$B_z = B_z$

$B_\rho = B\, a_\rho$

$= (y a_x - x a_y + z a_z) a_\rho$

$= y (a_x a_\rho) - x (a_y a_\rho) + z (a_z a_\rho)$

$= y \cos\phi - x (\sin\phi) + z(0)$

$\boxed{B_\rho = y \cos\phi - x \sin\phi}$

$B_\rho = B \cdot a_\rho$

$= (y a_x - x a_y + z a_z) \cdot a_\phi$

$= y(a_x \cdot a_\phi) - x(a_y \cdot a_\phi) + z(a_z \cdot a_\phi)$

$= y(-\sin\phi) - x(\cos\phi) + z(0)$

$$\boxed{B_\rho = -y\sin\phi - x\cos\phi}$$

$$\boxed{B_z = B_z}$$

$\theta = ?$ $\rho = ?$

$\theta = \tan^{-1}\left(\frac{y}{x}\right)$

$\rho = \sqrt{x^2 + y^2}$

$\rho = \sqrt{(-y)^2 + (-x)^2}$

$\rho = \sqrt{y^2 + x^2}$

As $x = \rho\cos\theta$

$y = \rho\sin\theta$

So $\theta = \tan^{-1}\left(\frac{x}{y}\right)$

29 August

Lecture 4

Spherical Coordinate System

r → The distance from origin to the point

Polar Angle(θ) → The angle between z axis and the line connecting to the origin.

Azimuthal Angle(ϕ) → The angle between the x axis and projection of the line connecting the origin to the point onto the xy plane

Represent longitudinal & latitudinal syst

RCS → x, y, z
CCS → ρ, ϕ, z
SCS → r, ϕ, θ

r (radius)

RCS → 3 plane
CCS → 2 plane, 1 curve
SCS → sphere, cone plane

Unit Vector

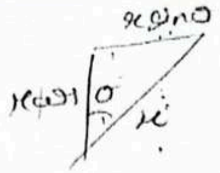
a_r, a_θ, a_ϕ

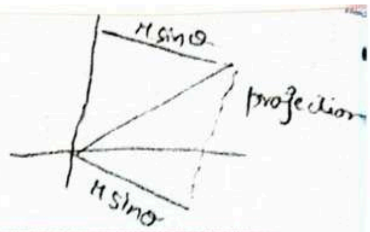
a_r → Radial unit vector, directed outward from the origin, normal to the sphere

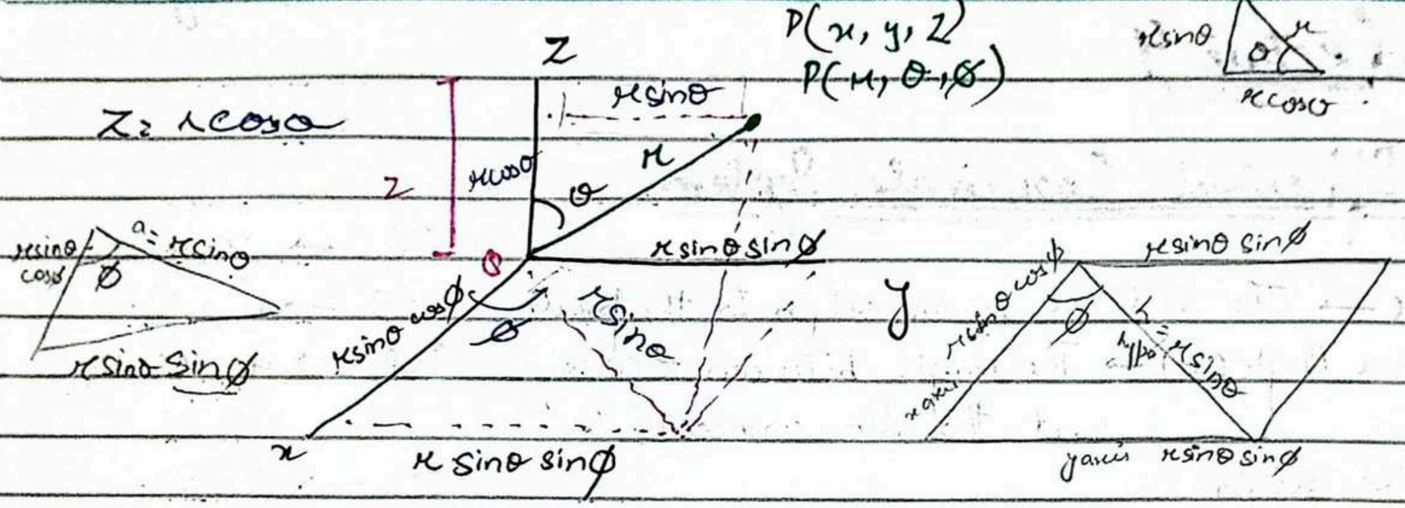
a_θ → Unit vector, tangent to the surface and pointing in the direction of increasing θ, normal to the ~~original~~ conical sphere

a_ϕ → unit vector tangent to the sphere and cone pointing in the direction of increasing ϕ

* unit vector show direction
* r vector ki direction ma a_r vector draw hoga
* Tagent radius sa ⊥ hota
* Sphere ka tagent par ya vector draw ga
* a_θ & a_ϕ are all tangent of sphere. They are 2 different vector representing different things in different direction
* These vectors are drawn at point of radius
* radius sa tagent draw hoga θ and ϕ ki direction ma







$$Z = r\cos\theta$$
$$x = r\sin\theta\cos\phi$$
$$y = r\sin\theta\sin\phi$$

* r (expressions ko Square ky kae common ley lm)

$$r = \sqrt{x^2+y^2+z^2}$$
$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$
$$(0 \le \phi \le 360^\circ)$$

derivation of ϕ

$$\frac{x}{y} = \frac{\cancel{r}\,\cancel{\sin\theta}\cos\phi}{\cancel{r}\,\cancel{\sin\theta}\sin\phi}$$
$$\frac{y}{x} = \frac{\sin\phi}{\cos\phi}$$
$$\frac{y}{x} = \tan\phi$$
$$\boxed{\phi = \tan^{-1}\left(\frac{y}{x}\right)}$$

$$z^2 = r^2\cos^2\theta$$
$$x^2 = r^2\sin^2\theta\cos^2\phi$$
$$y^2 = r^2\sin^2\theta\sin^2\phi$$
$$z^2+x^2+y^2 = r^2\cos^2\theta + r^2\sin^2\theta\cos^2\phi + r^2\sin^2\theta\sin^2\phi$$
$$z^2+x^2+y^2 = r^2(\cos^2\theta + \sin^2\theta\cos^2\phi + \sin^2\theta\sin^2\phi)$$
$$z^2+x^2+y^2 = r^2(\cos^2\theta + \sin^2\theta(\cos^2\phi + \sin^2\phi))$$
$$z^2+x^2+y^2 = r^2(\cos^2\theta + \sin^2\theta(1))$$
$$\boxed{r = \sqrt{x^2+y^2+z^2}}$$

Transformation of vector:

The transformation of vectors requires determining the dot products of spherical unit vectors with rectangular unit vector

Purpose of dot product: To find the component of a vector in the direction of a new unit vector (from SCS) we take the dot product of original vector's (from RCS) with the new vector.

Mathematical Representation:

Capital A_ϕ, A_θ, A_r are components

* Dot product magnitude and projection

$$A_{RCS} = A_x a_x + A_y a_y + A_z a_z$$

In spherical coordinate system

$$A_{SCS} = A_r a_r + A_\theta a_\theta + A_\phi a_\phi$$

The components of the vector A_{RCS} along the spherical coordinate axis

$$A_r = A_{RCS} \cdot a_r \qquad A_\theta = A_{RCS}\, a_\theta \qquad A_\phi = A_{RCS} \cdot a_\phi$$

$$A_r = A_{RCS} \cdot a_r$$

$$= (A_x a_x + A_y a_y + A_z a_z) a_r$$

$$= A_x(a_x a_r) + A_y(a_y a_r) + A_z(a_z a_r)$$

Dot products that gives the projection of the rectangular unit vectors onto the spherical unit vector 'a_r'

$$A_\theta = A_{RCS} \cdot a_\theta$$
$$= (A_x a_x + A_y a_y + A_z a_z) a_\theta$$
$$= A_x (a_x a_\theta) + A_y (a_y a_\theta) + A_z (a_z a_\theta)$$

$$A_\phi = A_{RCS} \cdot a_\phi$$
$$= (A_x a_x + A_y a_y + A_z a_z) a_\phi$$
$$= A_x (a_x a_\phi) + A_y (a_y a_\phi) + A_z (a_z a_\phi)$$

Lecture 5

3 Sep 2024

RCS — SCS

$$r = \sqrt{x^2+y^2+z^2}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\theta = \cos^{-1}\left(\frac{z}{\sqrt{x^2+y^2+z^2}}\right)$$

RCS $\longrightarrow$ CCS

$$\rho = \sqrt{x^2+y^2}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$

$$z = z$$

Transformation of Vectors

$$A_{RCS} = A_x a_x + A_y a_y + A_z a_z$$

$$A_{SCS} = A_r a_r + A_\theta a_\theta + A_\phi a_\phi$$

To find the component of a vector in the direction of new unit vector (from SCS) we take the dot product of the original vector's unit vector with the new vector

$A_r = A_{xcs} \cdot a_r$

$= (A_x a_x + A_y a_y + A_z a_z)(a_r)$

$= A_x(a_x a_r) + A_y(a_y a_r) + A_z(a_z a_r)$

a_r: points radially outward from the origin

a_θ: Tangent to the spherical surface and pointing in the direction of increasing θ

a_ϕ: Tangent to the circle of constant 'r' and 'θ' pointing the direction of increasing ϕ

$a_x a_r = a_r a_x = \cos\phi \sin\theta$

$a_y a_r = a_r a_y = \sin\theta \sin\phi$

$a_z a_r = a_r a_z = \cos\theta$

$A_r = A_x(a_x a_r) + A_y(a_y a_r) + A_z(a_z a_r)$

$$\boxed{A_r = A_x(\cos\phi \sin\theta) + A_y(\sin\theta \sin\phi) + A_z(\cos\theta)}$$

$A_\theta = A_{rcs}\, a_\theta$

$= (A_x a_x + A_y a_y + A_z a_z)\, a_\theta$

$= A_x(a_x a_\theta) + A_y(a_y a_\theta) + A_z(a_z a_\theta)$

$a_x a_\theta = a_\theta a_x = \cos\theta \cos\phi$

$a_y a_\theta = a_\theta a_y = \cos\theta \sin\phi$

$a_z a_\theta = a_\theta a_y = -\sin\theta$

$A_\theta = A_x(a_x a_\theta) + A_y(a_y a_\theta) + A_z(a_z a_\theta)$

$A_\theta = A_x(\cos\theta\cos\phi) + A_y(\cos\theta\sin\phi) - A_z(\sin\theta)$

$A_\phi = A_{rec}\, a_\phi$

$= (A_x a_x + A_y a_y + A_z a_z)\, a_\phi$

$$\boxed{A_\phi = A_x(a_x a_\phi) + A_y(a_y a_\phi) + A_z(a_z a_\phi)}$$

$a_x a_\phi = a_\phi a_x = -\sin\phi$

$a_y a_\phi = a_\phi a_y = +\cos\phi$

$a_z a_\phi = a_\phi a_z = 0.$

$A_\phi = A_x(a_x a_\phi) + A_y(a_y a_\phi) + A_z(a_z a_\phi)$

$$\boxed{A_\phi = -A_x\sin\phi + A_y\cos\phi}$$

* $a_z a_\phi = 0$ as ϕ donot lie in z plane it is in xy plane

* Moreover in ϕ there is no θ as it lies only in xy plane and θ is in z plane

$$A_{SCS} = \overbrace{A_r a_r + A_\theta a_\theta + A_\phi a_\phi}^{\text{Vector components}}$$

$$\boxed{\begin{aligned} A_{SCS} = & (A_x\cos\theta\sin\phi + A_y\sin\theta\sin\phi + A_z\cos\theta)\, a_r \\ & + (A_x(\cos\theta\cos\phi) + A_y(\cos\theta\sin\phi) - A_z\sin\theta)\, a_\theta + \\ & (-A_x\sin\phi + A_y\cos\phi)\, a_\phi \end{aligned}}$$

Q. $G = \frac{xz}{y} a_x \longrightarrow$ express it in spherical coordinate system

xyz are component

Step 1: $G = G_r a_r + G_\theta a_\theta + G_\phi$

$G_r = G_{rect} \cdot a_r$

$G_r = \frac{xz}{y} a_x a_r$

$G_r = \frac{(r\sin\theta\cos\phi)(r\cos\theta)}{r\sin\theta\sin\phi} (a_x a_r)$

$G_r = \left(\frac{r\cos\theta \cos\phi}{\sin\phi} \right) (a_x a_r)$

$G_r = \frac{r\cos\theta\cos\phi}{\sin\phi} (\sin\theta\cos\phi)$

$$\boxed{G_r = r\frac{\sin\theta\cos\theta\cos^2\phi}{\sin\phi}}$$

$G_\theta = G_{rect} \cdot a_\theta$

$= \left(\frac{xz}{y}\right) a_x a_\theta$

$= \frac{(r\sin\theta\cos\phi)(r\cos\theta)}{r\sin\theta\sin\phi} (a_x a_\theta)$

$= \frac{r\cos\phi\cos\theta}{\sin\phi} (\cos\theta\cos\phi)$

$$\boxed{G_\theta = r\frac{\cos^2\phi\cos^2\theta}{\sin\phi}}$$

$$G_\phi = G_{rect}\, a_\phi$$

$$= \frac{xz}{y}\, a_x a_\phi$$

$$= \frac{(r\sin\theta\cos\phi)(r\cos\theta)}{r\sin\theta\sin\phi}\,(a_x a_\theta)$$

$$= \frac{r\cos\theta\cos\phi}{\sin\phi}\,(a_x a_\theta)$$

$$= \frac{r\cos\theta\cos\phi}{\sin\phi}\,(-\sin\phi)$$

$$\boxed{G_\phi = r\cos\theta\cos\phi}$$

$$G = G_r a_r + G_\theta a_\theta + G_\phi a_\phi$$

$$G = \left(\frac{r\cos\theta\cos^2\phi\sin\theta}{\sin\phi}\right)a_r + \left(\frac{r\cos^2\theta\cos^2\phi}{\sin\phi}\right)a_\theta +$$

$$r\cos\theta\cos\phi\, a_\phi$$

Dot product of unit Vector

	a_r	a_θ	a_ϕ
a_x	$\sin\theta\cos\phi$	$\cos\theta\cos\phi$	$-\sin\phi$
a_y	$\sin\theta\sin\phi$	$\cos\theta\sin\phi$	$\cos\phi$
a_z	$\cos\theta$	$-\sin\theta$	0

Lecture 6:

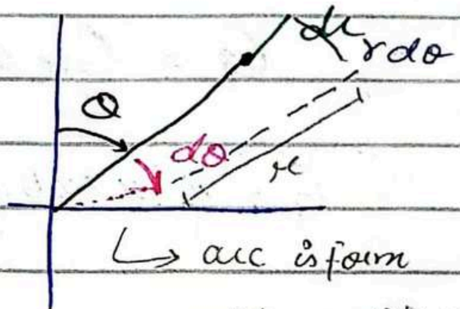
5 September

Differential elements in the spherical coordinate system

Differential Lengths

- Consider increasing r, θ and ϕ by small amounts $\rightarrow$ dr, $d\theta$ and $d\phi$ respectively

- $dr \rightarrow$ along the radial direction (r)
- $r d\theta \rightarrow$ along the polar direction (θ)
- $r \sin\theta d\phi \rightarrow$ along the azimuthal direction (ϕ)



↳ arc is form

we want to see distance

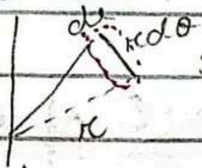
$l = r\theta$

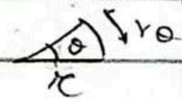
So we take $r d\theta$

Differential Surface Area:

- Differential Surface area spanned by r (dr) and θ $(r d\theta)$

  $\hookrightarrow = dr\,(r d\theta) = r\,dr\,d\theta$





Differential surface area spanned by r (dr) and ϕ $(r\sin\theta d\phi)$

$= r\,dr \sin\theta\, d\phi = r\sin\theta\, dr\, d\phi$

* Area simply need to (x) '2' lengths

* projection nechay girahay ga $r\sin\phi$ & $r\cos\phi$ me divide hoga hue yani ka across $r\sin\theta\, d\phi$ aayega ga.

Differential surface area spanned by θ $(r d\theta)$ and ϕ $(r\sin\theta d\phi)$

$= r^2 \sin\theta\, d\theta\, d\phi$

- Differential Volume $= dr \times r d\theta \times r\sin\theta d\phi$

$= r^2 \sin\theta \, dr \, d\theta \, d\phi$

$= dV$

For total volume we do integration.

$$\int_r dV = \int r^2 \sin\theta \, dr \, d\theta \, d\phi$$

$$V = \sin\theta \, d\theta \, d\phi \int r^2 dr$$

$$V = \sin\theta \, d\theta \, d\phi \times \frac{r^3}{3}$$

$$\boxed{V_r = \frac{r^3}{3} \sin\theta \, d\theta \, d\phi}$$

$$\int_\theta dV = \int r^2 \sin\theta \, dr \, d\theta \, d\phi$$

$$V_\theta = r^2 \, dr \, d\phi \int \sin\theta \, d\theta$$

$$\boxed{V_\theta = -r^2 \cos\theta \, dr \, d\theta}$$

$$\int_\phi dV = \int r^2 \sin\theta \, dr \, d\theta \, d\phi$$

$$V_\phi = r^2 \, dr \sin\theta \, d\theta \int d\phi$$

$$\boxed{V_\phi = r^2 \sin\theta \, \phi \, dr \, d\theta}$$

$$\int_\phi \int_\theta \int_r V = \int_\phi \int_\theta \int_r r^2 \sin\theta \, dr \, d\theta \, d\phi$$

$$= \int_\phi \int_\theta \sin\theta \, d\theta \, d\phi \int r^2 dr$$

$$= \int_\phi \int_\theta \sin\theta \, d\theta \, d\phi \times \frac{r^3}{3}$$

$$= -\cos\theta \frac{r^3}{3} \int d\phi$$

$$V = \frac{r^3}{3} \cos\theta \, \phi$$

1) $C(-3, 2, 1) \longrightarrow$ spherical coordinate

$r = \sqrt{x^2+y^2+z^2}$ $\quad \theta = \cos^{-1}\left(\frac{z}{\sqrt{x^2+y^2+z^2}}\right)$ $\quad \phi = \tan^{-1}\left(\frac{y}{x}\right)$

$r = \sqrt{3^2+2^2+1^2}$

$\boxed{r = 3.74}$ $\quad = \cos^{-1}\left(\frac{1}{3.74}\right)$ $\quad \phi = \tan^{-1}\left(\frac{2}{-3}\right)$

$\boxed{\theta = 74.49^\circ}$

$\boxed{\phi = 146.3^\circ}$

$C(r, \theta, \phi) = (3.74, 74.49, 146.3^\circ)$

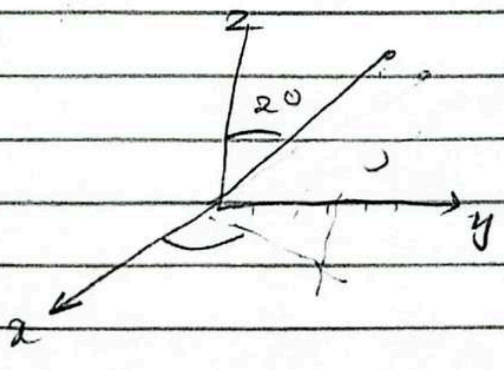
$D(r = 5, \theta = 20^\circ \ \phi = -70^\circ) \longrightarrow$ RCS

$x = r\sin\theta\cos\phi$ $\quad y = r\sin\theta\sin\phi$ $\quad z = r\cos\theta$

$x = 5\sin(20)\cos(-70)$ $\quad y = 5\sin(20)\sin(-70)$ $\quad z = 5\cos(20)$

$\boxed{x = 0.584}$ $\quad \boxed{y = -1.606}$ $\quad \boxed{z = 4.69}$

$D(x, y, z) = (0.584, -1.606, 4.69)$



12 September

Lecture 7

Columb's Law

Force between the two charged particles is $F = K\frac{Q_1 Q_2}{R^2}$

- Proportional to the magnitude of each charge (Q_1 and Q_2)
- Inversly proportional to the square of the distance b/w two charges (R)
- Acts along the line joining the two charges
- Its repulsive if the charges are of the same sign and attractive if they are of the opposite signs

- Q_1 and Q_2 are the magnitudes of the charges
- R is the distance b/w the charges
- k is the proportionality constant

$$k = \frac{1}{4\pi\varepsilon_0}$$

- ε_0 is the permittivity of free space

$$\varepsilon_0 = 8.854 \times 10^{-12}\ F/m$$

Components of columb's Law

1. Permittivity of free space (ε_0) Crucial for determining how electric fields interact in a vacuum. It is not dimensionless and has units $C^2/N.m^2$ or F/m

2. The columb (C) It is the SI unit electric charge

The charge of a single election is $-1.602 \times 10^{-19} C$

↳ due to election

1 coloumb corresponds to 6×10^{18} election

3 Force Magnitude: Force b/w two charges of 1C, separated by 1m in a vaccum

As $F = \frac{K Q_1 Q_2}{r^2}$

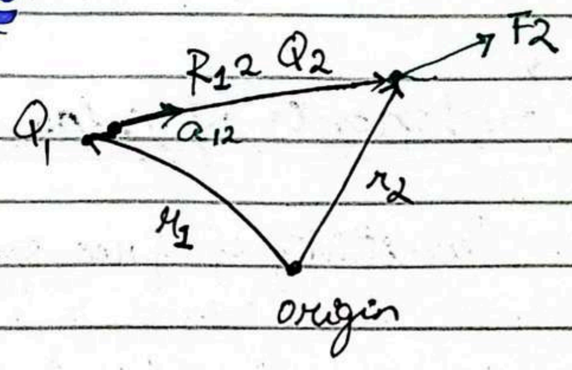
$$F = \frac{\frac{1}{4\pi\varepsilon_0} \times Q_1 Q_2}{r^2}$$

$$F = \frac{\frac{1}{4\pi \times 8.854 \times 10^{-12}} \times 1}{(1)^2}$$

$$\boxed{F = 8.987 \times 10^{9} N}$$

Vector form of Columb's Law

If Q_1 and Q_2 have like signs, the vector force F_2 on Q_2 is in the same direction as the vector R_{12}



$R_{12} = r_2 - r_1$

26 Sep 2024

Lecture 10

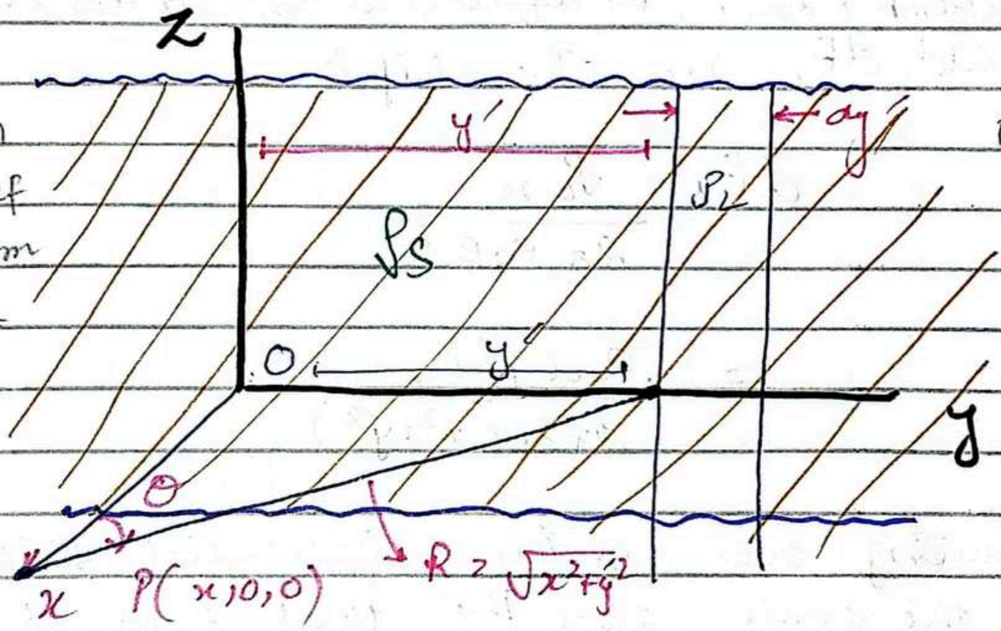
* If sheet of +ve charge than direction is outward
* If sheet of -ve charge than direction is inward

Field of a Sheet Charge

Surface Charge Density Amount of electric charge per unit area of a surface (C/m^2)

If thin sheet separated than it provides Line charge density

Any point in infinite sheet have uniform charges and electric field in this sheet will not vary



Infinit sheet of charge spread across yz plane where $\rho_S \rightarrow$ surface charge dens.

Infinite Sheet of charge Let's place an infinite of charge in the yz plane with uniform charge density

Due to the symmetry the electric field cannot vary along y axis or z axis because any point on the sheet looks identical along those directions

The line charge density of the thin strip (in infinite sheet of charge)

$$\rho_L = \rho_S \, dy'$$

$$R = \sqrt{x^2 + y'^2} \quad \text{(distance)}$$

Using Coloumb's Law, The differential contribution to The electric field dE_x from The strip is

$$dE_x = \frac{\rho_L x}{2\pi \varepsilon_0 R^2}$$

$$= \frac{\rho_S dy' x}{2\pi \varepsilon_0 (x^2 + y'^2)}$$

Integrating over all the differential strips along the y axis gives The total field at Ex at point 'p'

$$E_x = \int_{-\infty}^{\infty} dE_x$$

$$= \int_{-\infty}^{\infty} \frac{\rho_S dy' x}{2\pi \varepsilon_0 (x^2 + y'^2)}$$

$$= \frac{\rho_S x}{2\pi \varepsilon_0} \int_{-\infty}^{\infty} \frac{dy'}{x^2 + y'^2}$$

$\tan^{-1}(x) = \int \frac{1}{1+x^2}$

$\frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) = \int \frac{1}{x^2+a^2}$

$$= \frac{\rho_s x}{2\pi\varepsilon_0} \int_{-\infty}^{\infty} \frac{dy}{x^2+y^2}$$

$$= \frac{\rho_s x}{2\pi\varepsilon_0} \times \frac{1}{x} \tan^{-1}\left(\frac{y}{x}\right)\Big|_{-\infty}^{\infty}$$

$$= \frac{\rho_s x}{2\pi\varepsilon_0} \times \frac{1}{x}\left[\tan^{-1}\left(\frac{\infty}{x}\right) - \tan^{-1}\left(-\frac{\infty}{x}\right)\right]$$

$$= \frac{\rho_s x}{2\pi\varepsilon_0} \times \frac{1}{x}\left[\tan^{-1}(\infty) - \tan^{-1}(-\infty)\right]$$

$$= \frac{\rho_s}{2\pi\varepsilon_0}\left[\frac{\pi}{2} + \frac{\pi}{2}\right]$$

$$= \frac{\rho_s}{2\pi\varepsilon_0}[\pi]$$

$$\boxed{E_x = \frac{\rho_s}{2\varepsilon_0}}$$

$$\boxed{E_x = \frac{\rho_s}{2\varepsilon_0}\hat{a}_x}$$

Application of finite sheet of charge (Capacitor → 2 sheets work on the principle of electric field)

EM

Electric field

due to point charge

$$E = \frac{Q}{4\pi\varepsilon_0 R^2} \times \hat{a}_R$$

electric field vary across R^2

$R^2 \uparrow \quad E \downarrow$

due to Line charge

$$E = \frac{\rho_L}{2\pi\varepsilon_0 \rho} \times \hat{a}_\rho$$

Linearly decrease with distance

due to infinite sheet

$$E = \frac{\rho_s}{2\varepsilon_0} \hat{a}_n$$

ρ_s → surface charge density

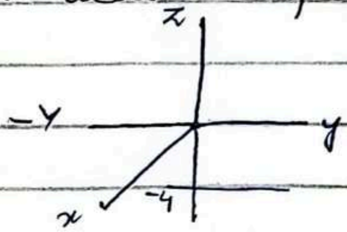
does not depend on distance

1 October 2024

Lecture 11

Sheet of Charges

Problem Three uniform sheet of charge are located in free space as follows



1) $3nC/m^2$ at $z=-4$
2) $6nC/m^2$ at $z=1$
3) $-8nC/m^2$ at $z=4$

Find E at point $P_A(2,5,-5)$ at point $P_C(-1,-5,2)$
at point $P_B(4,2,-3)$ at point $P_D(-2,4,5)$

$$E = \frac{\rho_s}{2\varepsilon_0}\hat{a_n}$$

$P_A = \sqrt{2^2+5^2+5^2}$

$P_A = 3\sqrt{6}$

at $3nC/m^2$ at $z=-4$ $P_A(2,5,-5)$

$$E = \frac{3\times10^{-9}}{2(8.85\times10^{-12})}$$

$$E_1 = 169.491\ V/m$$

at $6nC/m$

$$E = \frac{6\times10^{-9}}{2(8.85\times10^{-12})}$$

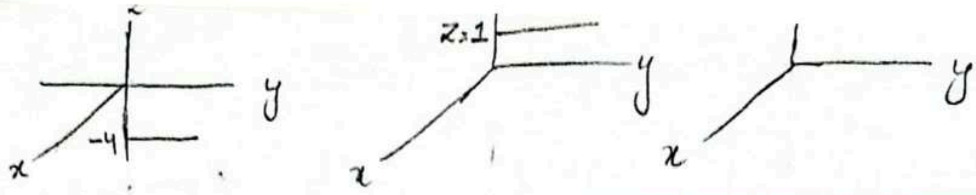
$$E_2 = 338.98\ V/m$$

at $-8nC/m^2$

$$E = \frac{-8\times10^{-9}}{2\times8.85\times10^{2}}$$

$$E_3 = -451.97\ V/m$$

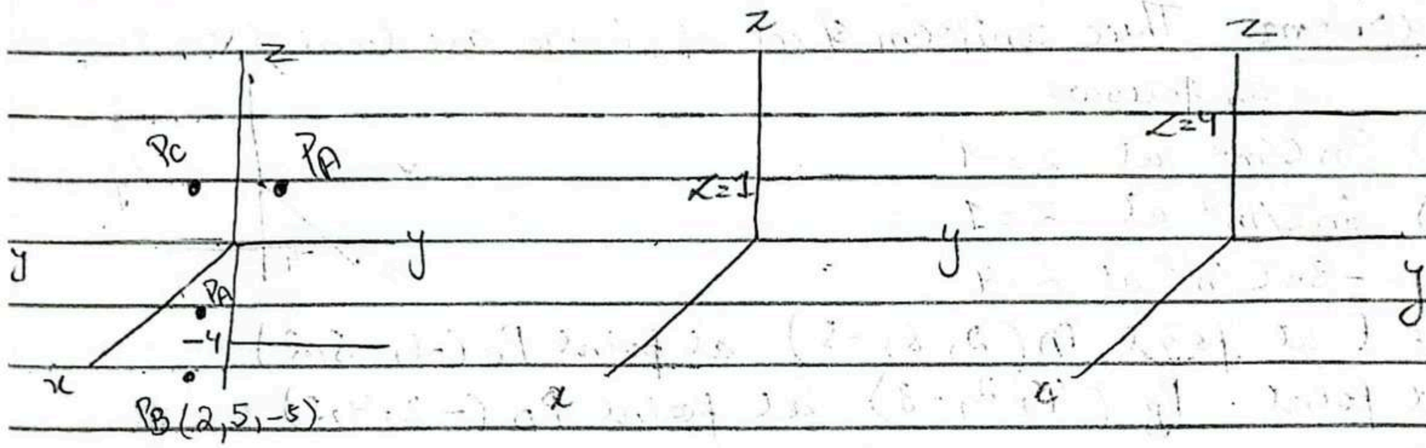
Problem 2.6

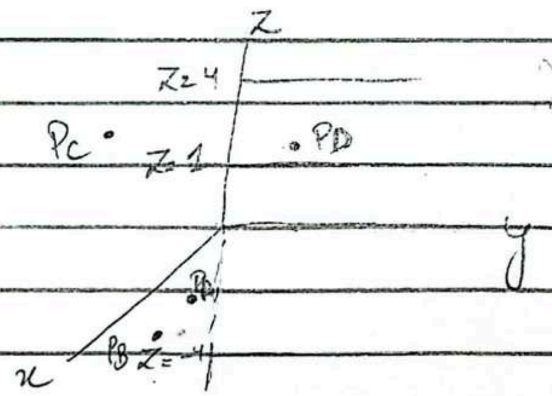


Total Electric field $\mathcal{E}_T = \mathcal{E}_1 + \mathcal{E}_2 + \mathcal{E}_3$

$= 169.491 + 338.91 - 451.97$

$\boxed{\mathcal{E}_T = 56.431 \text{ V/m}}$





30 oct 2024

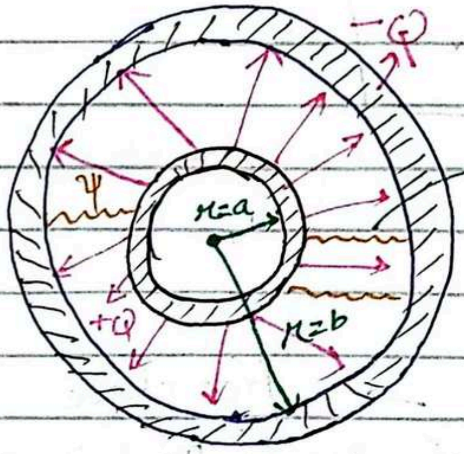
Lecture 12

Chapter 3

Electric flux Density

jo char charge ko reflect karta hai woo flux hai

outer surface par jo charge aya wo inner surface se reflected hai (opp) this is bcz of flux



This is different to magnetism

Displacement flux

. Electric flux (Ψ) is directly proportional to charge on the inner sphere

$$\Psi = Q$$

SI unit: Columb

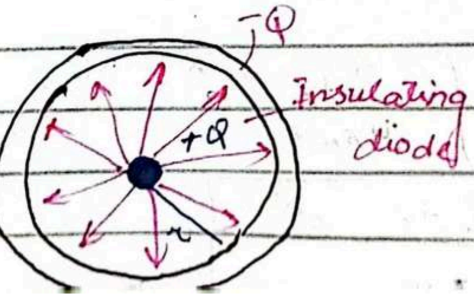
Electric flux density is define as the electric flux per unit area

unit (C/m^2)

- Electric flux at any given point is given by The no. of flux Lines crossing a surface normal to the Lines divided by the surface

\# distance changes here

\# field lines direction does not changes



inner sphere size is decrease so much that it act like a point charge

For point charge

Due to point charge, electric flux density

$$\boxed{D = \frac{\Psi}{4\pi r^2} \times \hat{a_r}}$$

it is surface area of sphere

as $\Psi = Q$

$$\boxed{D = \frac{Q}{4\pi r^2} \times \hat{a_r}}$$

$$E = \frac{Q}{4\pi \varepsilon_0 r^2}$$

$$\boxed{D = E\,\varepsilon_0} \longrightarrow \text{free space}$$

electric flux density due to point charge for free space

free space ← → air constant

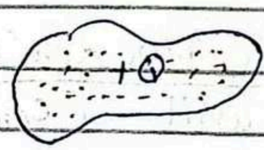
$$\boxed{D = E\,\varepsilon_0\,\varepsilon_r} \longrightarrow \text{for air}$$

Electric flux density due to point charge in free space

D

Koi bi surface in free space kisi dusri surface par flux find karna ho to aus surface ka flux sa find kar sakte

−Q



Volume charge density ↳ charge is evenly distributed on a surface

→ describe how much charge is distributed on small surface

$$E = \int_{vol} \frac{\rho_v\,dv}{4\pi \varepsilon_0 r^2} \times \hat{a_r} \quad (\text{by volume charge distribution})$$

$$E\,\varepsilon_0 = \int \frac{\rho_v\,dv}{4\pi r^2}\,\hat{a_r}$$

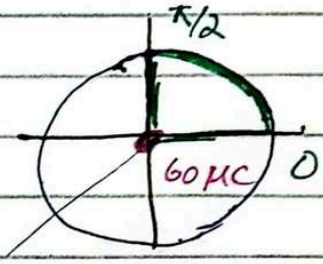
$$\boxed{D = \int \frac{\rho_v\,dv}{4\pi r^2} \times \hat{a_r}}$$

close surface
→ cylinder

Given a $60\mu C$ point charge located at the origin find the total electric flux passing through

a) The portion of the sphere $r = 26cm$ bounded by
$0 < \theta < \frac{\pi}{2}$ and $0 < \phi < \frac{\pi}{2}$

$\frac{1}{8}$ th section of the total sphere

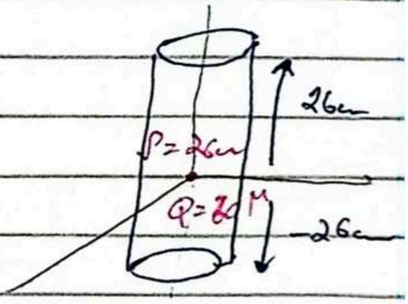


$\Psi = Q$

$\Psi = Q \times \frac{1}{8}$

$= 60 \times \frac{1}{8}$

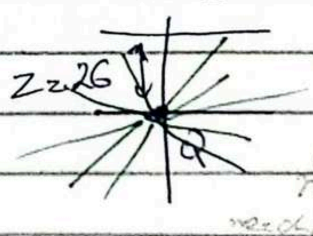
$\boxed{\Psi = 7.5\mu C}$



b) closed surface $\rho = 26cm$ $z = \pm 26cm$

$\Psi = 60\mu C$

Closed surface
andar 60μC
So overall 60μC



c) plane $z = 26cm$

$\Psi = \frac{1}{2} \times 60\mu C$

$\boxed{\Psi = 30\mu C}$

half
upper
jahan
ga half
neechay
neechay wala
hum consider rahe

Q. Calculate D in RCS at point P(2, -3, 6)

a) due to point charge $Q_A = 55$ mC at Q(-2, 3, -6) point charge

$= QP = P - Q$

$r = r - r' = (2, -3, 6) - (-2, 3, -6)$

$4, -6, 12$

$|r - r'| = \sqrt{4^2 + 6^2 + 12^2}$

$\boxed{r = 14}$

$$\hat{r} = \frac{4}{14}\hat{a_x} - \frac{6}{14}\hat{a_y} + \frac{12}{14}\hat{a_y}$$

$$= \frac{2}{7}a_x - \frac{3}{7}\hat{a_y} + \frac{6}{7}\hat{a_z}$$

$$D = \frac{55 \times 10^{-3}}{4\pi(14)^2}\left(\frac{2}{7}a_x - \frac{3}{7}a_y + \frac{6}{7}a_z\right)$$

$$= 2.23 \times 10^{-5}\left(\frac{2}{7}a_x - \frac{3}{7}a_y + \frac{6}{7}a_z\right)$$

$$= 6.38 \times 10^{-6}\hat{a_x} - 9.55 \times 10^{-6}\hat{a_y} + 1.91 \times 10^{-5}\hat{a_z}$$

$$\boxed{D = 6.38\hat{a_x} - 9.55\hat{a_y} + 19.1\hat{a_z}\ \mu C/m^2}$$

CamScanner

10 Oct 2024

Lecture 14

Applications of Gauss's Law: Symmetrical Charge distributions

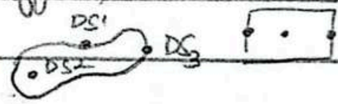
$$\oint D_s \cdot ds = Q$$ → only valid/applicable for symmetrical surfaces

total electric flux through a closed surface is proportional to the charge enclosed

Symmetry:

Spherical, Cylinderical, planar symmetrical

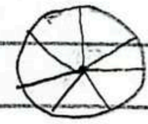
* if surface is not symmetrical than Ds is not same. Ds at every point is different



Key Conditions for Applying Gauss's Law

To efficiently use Gauss's Law, we must choose a gussian surface (closed surface that enclose the charge and is symmetrical) that satisfies the following conditions

* Symmetrical surface Ds same everywhere



1. D (Electric flux density) must be either normal or tangential to the surface
   - if D is normal to the surface
     $D \cdot ds = D ds$
   - if D is tangential to the surface
     $D \cdot ds = 0$

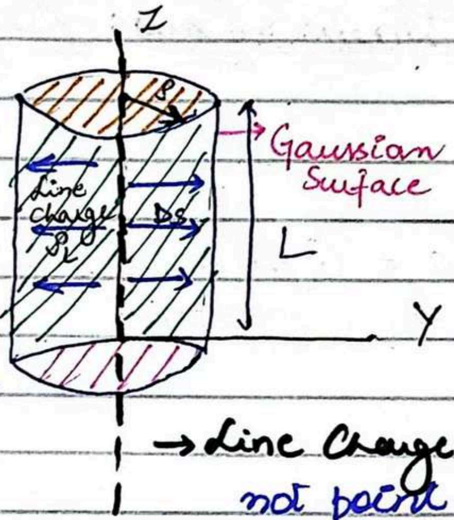
* coloumbs Law can be applied on any surface

2. The magnitude of D must be constant over the surface where it is normal (This allows us to pull Ds outside of the integral)

Leaving only the surface area inside the integral)
(Ds must be symmetrical also)

For Line Charge

Gaussian surface of an infinite Line charge
D is constant in magnitude wherever perpendicular to the cylindrical surface



Electric flux can be noted at
- Sides
- Top
- bottom

→ Line Charge (radially outward not point toward upward & downward direction) so we take zero over there)

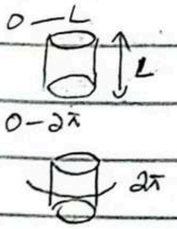
$$Q = \oint Ds\, ds$$

$$= Ds \int ds$$

$$= Ds \int_{Top} ds + Ds \int_{bottom} ds + Ds \int_{sides} ds$$

$$= 0 \int_{Top} ds + 0 \int_{bottom} ds + Ds \int_{sides} ds$$

here integral limit
0—L



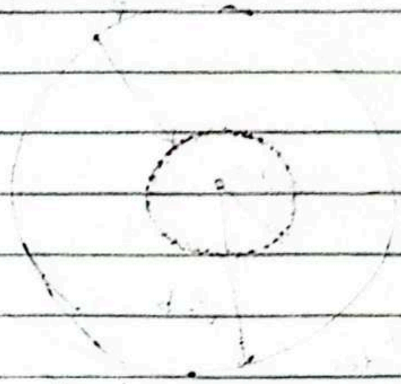
$$= Ds \int_{z=0}^{L} \int_{\phi=0}^{\phi=2\pi} \rho\, d\phi\, dz$$

↳ radius

$$= Ds\, 2\pi \rho L$$ (sides area in which top & bottom is excluded)

$$Q = D_S \, 2\pi \rho L$$

$$D_S = \frac{Q}{2\pi \rho L}$$



$$D_S = D_\rho = \frac{Q}{2\pi \rho L}$$

Also for line charge

$$Q = \rho_L L$$

then

$$D_S = D_\rho = \frac{\rho_L \cancel{L}}{2\pi \rho \cancel{L}}$$

$$\boxed{D_S = D_\rho = \frac{\rho_L}{2\pi \rho}}$$

than electric field

$$\boxed{E = \frac{\rho_L}{2\pi \varepsilon_0 \rho}}$$ because $D = E\varepsilon_0$

For Point Charge

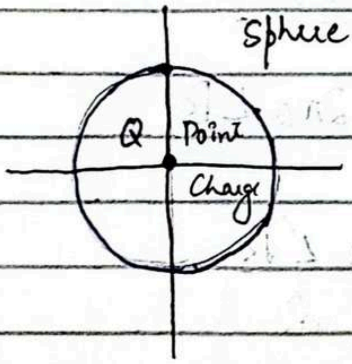
For symmetrical surface columb's law and gauss's law are consistent

$$Q = \oint D_S \, ds$$

$$Q = D_S \oint ds$$

$$Q = D_S (4\pi r^2)$$

$$\boxed{D_S = \frac{Q}{4\pi r^2} \hat{a_r}}$$

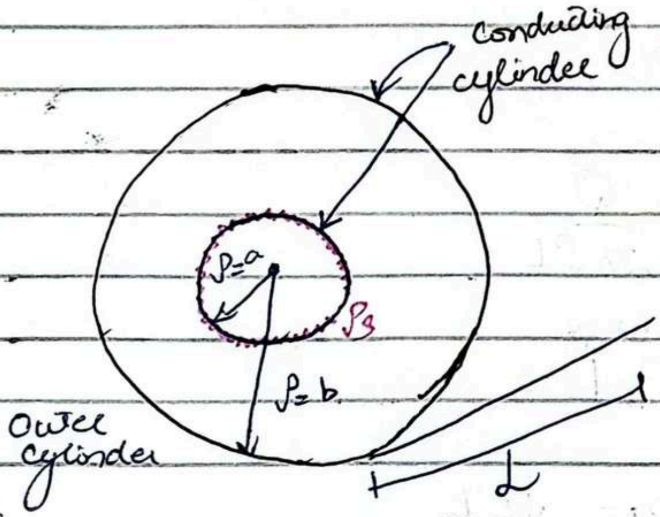


$$\boxed{E = \frac{Q}{4\pi \varepsilon_0 r^2} \hat{a_r}}$$

Coaxial Cable with cylinderical Symmetry:

$\rho_s \rightarrow$ inner cylinder surface charge density

as in Line charge there is cylinder which is also a conductor so they can corelated but in this case it has surface charge density instead of line charge



* Line charge imaginary cylinder drawn as gaussian surface to apply gauss's Law here also an imaginary cylinder of radius ρ this is done to interrelate both formulas

* if $\rho > b$ (as charge is finished on b so outside wire which is due to inside wire on cylinder there is no charge)

$a < \rho < b$ ↳ imaginary cylinder radius

Formula of Line charge $\quad D_S = D_\rho = \dfrac{Q}{2\pi\rho L} \Rightarrow \boxed{Q = D_S 2\pi\rho L}$

The total charge on a Length L is of the inner conductor is

$$Q_{inner} = \int_{z=0}^{L}\int_{\phi=0}^{2\pi} \rho_s \, a \, d\phi \, dz$$

↳ radius of inner conductor

$$\boxed{Q = 2\pi a L \rho_s}$$

$Q = D_s 2\pi\rho L \qquad Q = 2\pi a L \rho_s$

equating both

$D_s \cancel{2\pi} \rho \cancel{L} = \cancel{2\pi} a \cancel{L} \rho_s$

$$\boxed{D_s = \frac{a\rho_s}{\rho}\hat{a}_\rho}$$

→ radius of wire varying with different condition

$Q_{\substack{inner \\ enclosed}} = 2\pi a L \rho_s$

according to Faraday's law charge enclose inner is same but negative to charge enclose outer

$Q_{enclose\ outer} = -Q_{enclosed\ inner}$

$= -2\pi a L \rho_s$

$Q_{enclose\ outer} = -Q_{enclosed\ inner}$

$\cancel{2\pi} b \cancel{L} \rho_{s\ outer} = \cancel{2\pi} a \cancel{L} \rho_{s\ inner}$

$\dfrac{\rho_{s\ outer}}{\rho_{s\ inner}} = \dfrac{-a}{b}$

$$D = (0.32\times10^{-6} + 4\pi r^2 (Q))a_r$$

$$\rho_s = \frac{-0.32\times10^{-6}}{4\pi(3\times10^{-2})^2}$$

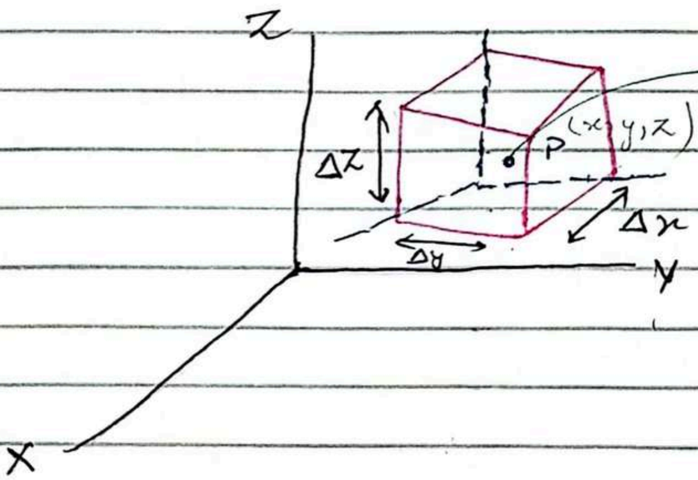
$$\boxed{\rho_s = -28.3\ \frac{\mu C}{m^2}}$$

Lecture 15

15 Oct 2024

Application of Gauss's Law : Differential Volume element

Small volume is taken as surface is not symmetrical



$D = D_0 = D_{x_0} a_x + D_{y_0} a_y + D_{z_0} a_z$

- For symmetrical surfaces, electric field or electric flux is constant (Gaussian Surface)
- What if the surface is not symmetrical → We will consider a very small differential volume element so that Gauss's Law can be applied on it to determine electric flux.

Gauss's Law

$$\oint D \cdot ds$$

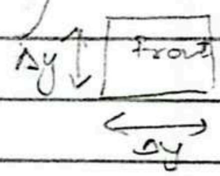
$$= \int_{\text{front}} D \cdot ds + \int_{\text{Back}} D \cdot ds + \int_{\text{right}} D \cdot ds + \int_{\text{left}} D \cdot ds + \int_{\text{Top}} D \cdot ds + \int_{\text{Bottom}} D \cdot ds$$

$S = L \times B$

$S = \Delta y \Delta z$

$$\int_{\text{front}} D\,ds = D_{front} \cdot \Delta S_{front}$$ (direction of electric flux is outward which is x axis)

$$= D_{front}\, \Delta y \Delta z\, a_x$$



Now to find D_{front} (D_x at the front surface)

The front face is at a distance of $\frac{\Delta x}{2}$ from point P

point P is at center. it is not covering distance Δx as at center so distance is $\frac{\Delta x}{2}$

$$D_{front} = D_{x_0} + \frac{\Delta x}{2} \times \text{Rate of change } D_x \text{ with } x$$

bcz D_x changing w.r.t to distance

$$D_{front} = D_{x_0} + \frac{\Delta x}{2} \frac{\partial D_x}{\partial x}$$

D_{x_0} pura cube ma kahi bi a rata hai initially

where D_{x_0} is value of D_x at P

and partial derivative is used to express the rate of change of D_x with x

$$\boxed{D_{front} = \left(D_{x_0} + \frac{\Delta x}{2} \frac{\partial D_x}{\partial x}\right) \Delta y \Delta z\, \hat{a_x}}$$

$$\int_{\text{back}} D \cdot ds = D_{Back} \cdot \Delta S_{back} = D_{s_{back}} (\Delta y \Delta z (-\hat{a_x}))$$

$$= -D_{s_{back}} \Delta y \Delta z$$

$$D_{Back} = D_{x_0} - \frac{\Delta x}{2} \frac{\partial D_x}{\partial x}$$

$$\int_{Back} D \cdot ds = -\left(D_{x0} - \frac{\Delta x}{2} \frac{\partial D_x}{\partial x} \right) \Delta y \Delta z$$

$$\boxed{\int_{Back} D \cdot ds = \left(-D_{x0} + \frac{\Delta x}{2} \frac{\partial D_x}{\partial x} \right) \Delta y \Delta z}$$

as along all x axis

$$= D_{front} + D_{Back}$$

$$= \left(D_{x0} + \frac{\Delta x}{2} \frac{\partial D_x}{\partial x} \right) \Delta y \Delta z + \left(-D_{x0} + \frac{\Delta x}{2} \frac{\partial D_x}{\partial x} \right) \Delta y \Delta z$$

$$= \Delta y \Delta z \left(\cancel{D_{x0}} + \frac{\Delta x}{2} \frac{\partial D_x}{\partial x_1} - \cancel{D_{x0}} + \frac{\Delta x}{2} \frac{\partial D_x}{\partial x} \right.$$

$$= \left(\frac{\Delta x}{2} + \frac{\Delta x}{2} \right) \frac{\partial D_x}{\partial x} \Delta y \Delta z$$

$$\boxed{\oint_{x\,axis} D\, ds \quad \frac{\partial D_x}{\partial x} \Delta x \Delta y \Delta z}$$

initial vector ho hum na find kiya tha

along all y axis (right & left)

$$\boxed{\oint_{y\,axis} D\, ds = \frac{\partial D_y}{\partial y} \Delta x \Delta y \Delta z}$$

along all z axis (top & bottom)

$$\boxed{\oint_{z\,axis} D\, ds = \frac{\partial D_z}{\partial z} \Delta x \Delta y \Delta z}$$

Example - Problem D3.6

Total D ↦ D across charge enclose in surface

$$D = D_x + D_y + D_z$$

$$D = \frac{\partial D_x}{\partial x} \Delta x \Delta y \Delta z + \frac{\partial D_y}{\partial y} \Delta x \Delta y \Delta z + \frac{\partial D_z}{\partial z} \Delta x \Delta y \Delta z$$

$$\oint_{total} D\,ds = \left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} \right) \underbrace{\Delta x \Delta y \Delta z}_{\Delta V}$$

→ vector quantity

As $Q_{enclosed} = \oint_{total} D \cdot ds$] Gauss Law

$$Q_{enclosed} = \underbrace{\left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} \right)}_{\text{Divergence}} \Delta V$$

$$Q_{encl} = \nabla D \cdot \Delta V$$

Divergence ← → electric flux density

Charge enclose per unit volume [$\frac{Q_{encl}}{\Delta V} = \nabla D$

$$\boxed{\rho_V = \nabla D}$$

↳ volume charge density

17 Oct 2024

Lecture 16

Divergence Theorem

The del operator (∇) is a vector operator used to perform several important operations in vector calculus including divergence, gradient and curl

$$\nabla = \frac{\partial}{\partial x}\hat{a_x} + \frac{\partial}{\partial y}\hat{a_y} + \frac{\partial}{\partial z}\hat{a_z}$$

- $\hat{a_x}$, $\hat{a_y}$ and $\hat{a_z}$ are unit vectors in the x, y and z direction
- Partial derivatives indicate rates of change in the respective direction

The del operator behaves like a vector but instead of returning scalars it return partial derivatives when applied to scalar and vector fields

1. Gradient (∇V) → applied to scalar field 'V' — yielding a vector
2. Divergence ($\underset{\text{vector}}{\nabla}\cdot \underset{\text{vector}}{D}$) → applied to a vector field — yielding a scalar
3. Curl ($\nabla \times D$) → applied to a vector — yielding another vector

Divergence of a vector field:

Divergence represents the rate at which the vector field is 'spreading out' from a point

If we take

$$D = D_x\hat{a_x} + D_y\hat{a_y} + D_z\hat{a_z}$$

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then

$$\nabla \cdot D = \frac{\partial (D_x)}{\partial x}\hat{a_x} + \frac{\partial (D_y)}{\partial y}\hat{a_y} + \frac{\partial (D_z)}{\partial z}\hat{a_z}$$

$$\nabla = \left(\frac{\partial}{\partial x}\hat{a_x} + \frac{\partial}{\partial y}\hat{a_y} + \frac{\partial}{\partial z}\hat{a_z}\right)$$

$$D = D_x\hat{a_x} + D_y\hat{a_y} + D_z\hat{a_z}$$

$$\nabla \cdot D = \left(\frac{\partial}{\partial x}\hat{a_x} + \frac{\partial}{\partial y}\hat{a_y} + \frac{\partial}{\partial z}\hat{a_z}\right)\left(D_x\hat{a_x} + D_y a_y + D_z a_z\right)$$

$$= \frac{\partial}{\partial x}D_x(\hat{a_x}\cdot\hat{a_x}) + \frac{\partial}{\partial x}D_y(a_x \cdot a_y) + \frac{\partial}{\partial x}D_z(a_x\, a_z) +$$

$$\frac{\partial}{\partial y}D_x(\hat{a_x}\cdot\hat{a_y}) + \frac{\partial}{\partial y}D_y(a_y\, a_y) + \frac{\partial}{\partial y}(D_z)(a_y\, a_z) +$$

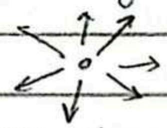
$$\frac{\partial}{\partial z}D_x(a_z\, a_x) + \frac{\partial}{\partial z}D$$

The divergence of a vector field at any field represents that whether the field is acting as a source (positive Divergence) or as sink (negative divergence → converging) at the point

If coming to point

−ve charge

if spreading out



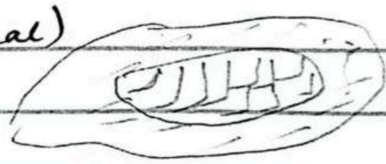
+ve charge

Divergence theorem connects the behaviour of a vector field inside a volume to the behaviour of the field on the surface enclosing that volume

Mathematically

$$\int_S D\,ds = \int_{vol} (\nabla\cdot D)\,dv$$

- $\int D\cdot ds$ represent the flux through the surface S enclosing the volume
  ↳ (double integral)
- $\int_{vol} (\nabla D)\,dv$ is the integral of divergence D throughout the volume
  ↳ (Triple integral)



The Divergence theorem state that the total flux existing a close surface is equal to the integral of the divergence of the field within a volume enclosed by the surface

Divergence in different coordinate system

In Cylinderical Coordinate System :

$$\nabla\cdot D = \frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho D_\rho) + \frac{1}{\rho}\frac{\partial D_\phi}{\partial\phi} + \frac{\partial D_z}{\partial z}$$

Gauss's Law

$$\int D \cdot ds = Q_{enclosed}$$

$$\int D \cdot ds = \int_{vol} (\nabla D) dV$$

$\rho_V = \nabla D$

$$\int D \cdot ds = Q_{enclosed} = \int_{vol} (\nabla D) dV$$

Keyword
div D
↳ specific divergence

Divergence and Maxwell's Equation

electric flux density divergence (very small area)

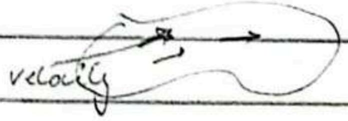
$$\frac{\partial}{\partial x}(D_x) + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} = \lim_{\Delta V \to 0} \frac{\oint D \cdot ds}{\Delta V} = \lim_{\Delta V \to 0} \frac{Q_{enc}}{\Delta V}$$

$$= \rho_V$$

$$\frac{\partial (A_x)}{\partial x} + \frac{\partial (A_y)}{\partial y} + \frac{\partial A_z}{\partial z} = \lim_{\Delta V \to 0} \frac{\int A \cdot ds}{\Delta V}$$ → A could be velocity force or any vector quantity

→ can be any vector source, displacement

$$\boxed{\text{Divergence of } A = \text{div} A = \lim_{\Delta V \to 0} \frac{\oint A \cdot ds}{\Delta V}}$$ → Maxwell's Equation



SCS → Spherical coordinate system

$$\operatorname{div} D = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 D_r\right) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\, D_\theta) + \frac{1}{r\sin\theta}\frac{\partial D_\phi}{\partial\phi}$$

D3-6 , D3-

SCS → Spherical Coordinate system

$$\operatorname{div} D = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 D_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\, D_\theta) + \frac{1}{r\sin\theta}\frac{\partial D_\phi}{\partial\phi}$$

Lecture 17

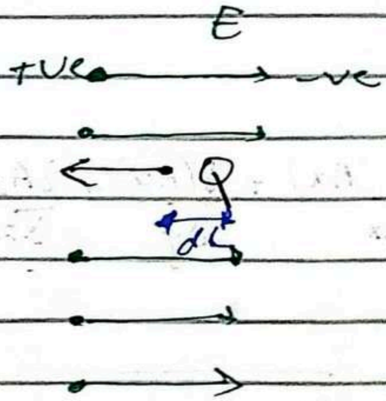
22 oct 2024

Topic 4.1

Energy Expended In Moving a charge in an Electric Field

$F_E = QE \longrightarrow$ +ve to −ve

Force exerted by the field



Force to move the charge against the Field

$$F_{app} = -QE$$

↳ Force is a vector quantity (so direction represented)

Work done in Moving a charge

Moving the charge Q in a small distance dl written an electric field E, the component of the force in the direction of dL is:

we find force applied on small length dL as it will be same on whole surface or on small part as magnitude is same

$$F_L = FE \cdot \hat{L} = QE \cdot \hat{L}$$

↳ Unit vector in the direction of dl

$\hat{L}$ is a unit vector in the direction of dL

$$F_{applied} = -QE \cdot \hat{L}$$

The differential work done by the applied in moving the charge by a small distance dL is given by

$$dW = F_{app} \cdot dL$$
$$dW = -QE \cdot dL$$

* Q showing magnitude no direction E·dl ka dot product so E and dl is perpendicular

Conditions when work done is zero

When E and dL are perpendicular

When $Q = 0$

if $dL = 0$

if $E = 0$

work done depend on
* Electric field
* path dL

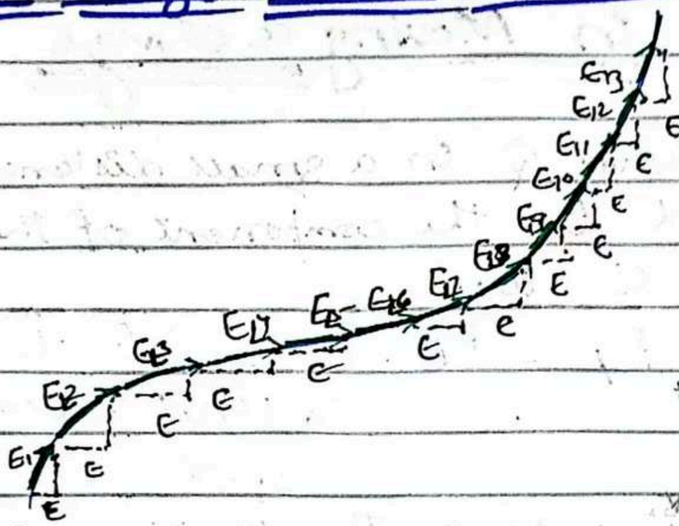
In moving the charge by a

$$W = -Q \int_{initial}^{final} E \cdot dL$$

work done over a finite distance

same through

Topic 4.2 Line Integral in a Uniform Electric field



* E is same every where
These all are component
* We have to move charge from A to B
* Electric field is uniform

$$\boxed{W = -Q \int_{initial}^{final} E \cdot dl}$$

Assuming uniform electric field consider the path between A and B. We want to calculate the work done in moving a charge point Q along its path

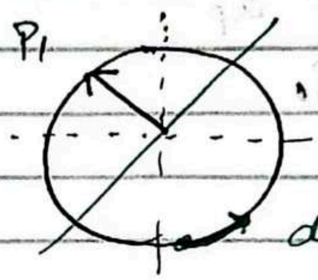
$$W = -Q \int_B^A E \cdot dL$$

$$W = -QE \int_B^A dL \implies \int_B^A dL = L_{BA}$$

$$\boxed{W = -QE \cdot L_{BA}}$$

This means the work done in a uniform electric field depends only on L_{BA} (path from A to B)

Line Integral In a uniform electric field (Circular path)



Infinite Line charge ρ_L
(Z axis) (mean outward)

$dL = \rho_1 d\phi\, a\phi$

$$W = -Q\int_0^{2\pi} E \cdot dl$$

$$E = E_\rho \hat{a_\rho} = \frac{\rho_L}{2\pi \varepsilon_0 \rho_1} \hat{a_\rho}$$

circular path distance 0 to 2π

$$dL = \rho_1 d\phi\, a\phi$$

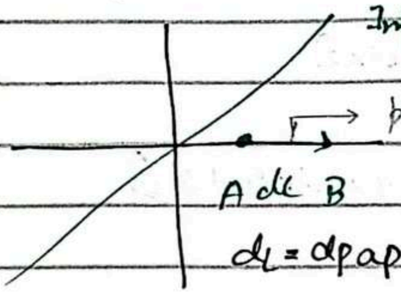
$$W = -Q\int_0^{2\pi} E \cdot dl$$

$$= -Q\int_0^{2\pi} \frac{\rho_L}{2\pi \varepsilon_0 \rho_1} \hat{a_\rho} \cdot \rho_1 d\phi\, a\phi$$

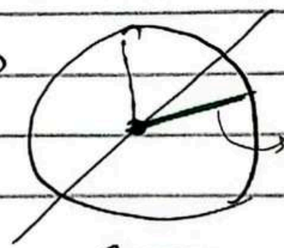
$$= -Q\int_0^{2\pi} \frac{\rho_L}{2\pi \varepsilon_0} d\phi\, \underbrace{\hat{a_\rho}\, \hat{a\phi}}_{}$$

perpendicular dot product zero

$$\boxed{W = 0}$$



Now we want to move outward
we want to move in direction of radius



$S_2 r O$

$$E = E_\rho \hat{a}_\rho = \frac{\rho_L}{2\pi\varepsilon_0 \rho} \hat{a}_\rho$$

$$dl = d\rho\, \hat{a}_\rho$$

$$W = -Q\int_A^B E\,dl$$

$$= -Q\int_A^B \frac{\rho_L}{2\pi\varepsilon_0\rho}\,\hat{a}_\rho \, d\rho\, \hat{a}_\rho$$

$$= -Q\int_A^B \frac{\rho_L}{2\pi\varepsilon_0}\,\frac{d\rho}{\rho}\,\underbrace{\hat{a}_\rho\hat{a}_\rho}$$

parallel to each other dot product = 1

$$= \frac{-Q\,\rho_L}{2\pi\varepsilon_0}\int_A^B\frac{d\rho}{\rho}\,(1)$$

$$\int \frac{1}{\rho}\,d\rho = \ln|\rho|\Big|_A^B = \ln|B| - \ln|A| = \ln|B-A| = \ln\left|\frac{B}{A}\right|$$

$$\boxed{W = \frac{-Q}{2\pi\varepsilon_0}\,\rho_L \ln\left(\frac{B}{A}\right)} \rightarrow \text{radially moving}$$

Line integrals in different coordinate system

RCS : $dl = dx\,\hat{x} + dy\,\hat{y} + dz\,\hat{z}$

CCS : $dL = d\rho\,\hat{\rho} + \rho\, d\phi\,\hat{\phi} + dz\,\hat{z}$

SCS $dL = dr\,\hat{r} + r\,d\theta\,\hat{\theta} + r\sin\theta\, d\phi\,\hat{\phi}$

Example 7.1, 4.2

24 oct 2024

Lecture 18

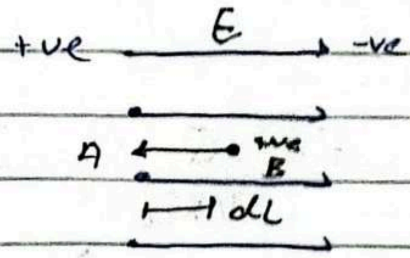
Topic : 4-3

Potential Difference and Potential

Potential difference is the work done by an external force to move a unit positive charge from one point to another in an electric field.

$$W = -Q\int E \cdot dl$$

(W in J, Q in C)



$$V = \frac{W}{Q} = \frac{J}{C}$$

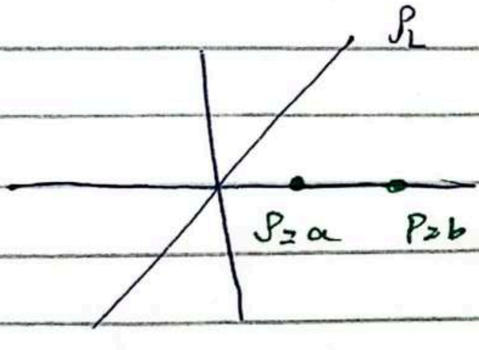
$$\boxed{V = -\int_{\text{initial}}^{\text{final}} E \cdot dl}$$

$$V = -\int_{A}^{A} E \cdot dl$$

(we are moving against electric field)

units $J/C \longrightarrow$ volts

Example

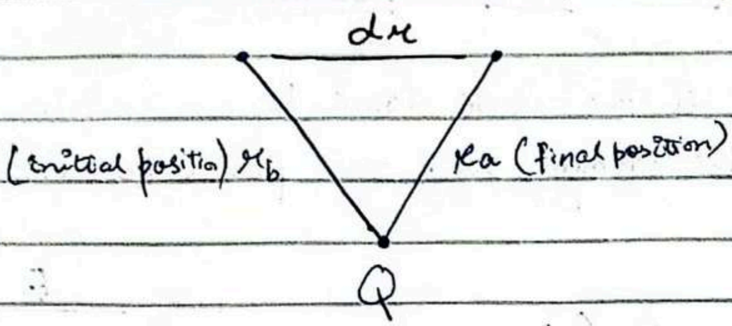


$$E = \frac{\rho_L}{2\pi\varepsilon_0\rho}\,\hat{a}_\rho$$

$$W = \frac{Q\rho_L}{2\pi\varepsilon_0}\ln\left(\frac{b}{a}\right)$$ (work done from b to A)

$$\frac{W}{Q} = V_{ab} = \frac{\rho_L}{2\pi\varepsilon_0}\ln\left(\frac{b}{a}\right)$$

Potential Difference from a point charge



Main components of potential difference

$V = \int E \cdot dl$
↳ E
↳ dl

$$E = \frac{Q}{4\pi\varepsilon_0 r^2}\hat{a_r}$$

$$dl = dr \cdot \hat{a_r}$$

$$V_{AB} = -\int E \cdot dl$$

$$= -\int_{r_b}^{r_a} \frac{Q}{4\pi\varepsilon_0 r^2}\, dr \quad \hat{a_r}\cdot\hat{a_r}$$

$$= -\frac{Q}{4\pi\varepsilon_0}\int_{r_b}^{r_a} \frac{1}{r^2}\, dr$$

$$= -\frac{Q}{4\pi\varepsilon_0}\int_{r_b}^{r_a} r^{-2}\, dr$$

$$= -\frac{Q}{4\pi\varepsilon_0}\left.\frac{r^{-1}}{-1}\right|_{r_b}^{r_a}$$

$$= +\frac{Q}{4\pi\varepsilon_0}\left.\frac{1}{r}\right|_{r_b}^{r_a} \qquad \frac{1}{r_a} - \frac{1}{r_b}$$

$$V_{AB} = \frac{Q}{4\pi\varepsilon_0}\frac{1}{(r_a - r_b)}$$

$$V_{AB} = \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{r_a} - \frac{1}{r_b}\right)$$

Absolute potential

↳ reference point ground

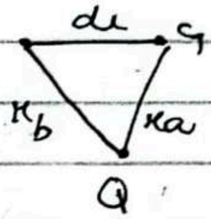
spherical and cylindrical (taken zero reference)

potential at infinity point is zero

Absolute potential $V = -\int E\,dl$

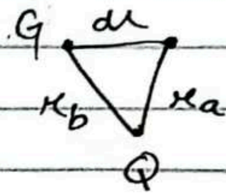
(w.r.t ground)

If r_a is at infinity (grounded)



$$V = \frac{Q}{4\pi\varepsilon_0}\left[-\frac{1}{r_B}\right]$$

If r_b is at grounded



$$V = \frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{r_a}\right]$$

$$\boxed{V = \frac{Q}{4\pi\varepsilon_0 r}}$$

→ If here r is ∞ $V = 0$

→ as r decrease potential increases

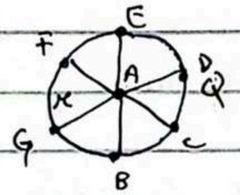
→ potential workdone by external force to move a charge from one point to another

→ as r decreases we need more work done or energy to move the charge to that point

Equipotential Surfaces

→ Consists of all the points in the space that have the same value of potential

→ No work is required to move a charge along an equipotential surface because the potential difference b/w two points on the surface is zero



* potential difference b/w A to B is V and potential difference b/w A to C is V so potential difference b/w B to C is zero
* On surface w.r.t point A sab ka potential same hai

$$E = \frac{Q}{4\pi\varepsilon_0 r^2}\,\hat{a_r} \text{ (due to point charge)}$$

$$V = \frac{Q}{4\pi\varepsilon_0 r}$$

$V \propto \frac{1}{r}$ — Energy → less influence

$E \propto \frac{1}{r^2}$ (strongly influence by distance) — Force → more influence → $F = q_1 E$

Potential difference is scalar quantity

Potential difference of a system of charges and the conservative property

Potential due to a single point Q_1 located at position r_1

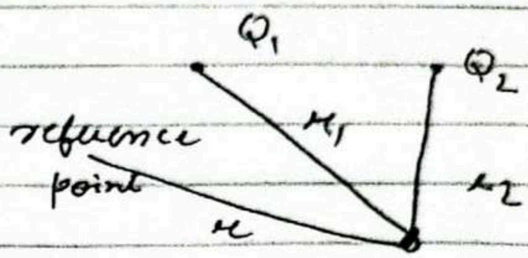
$$\boxed{V(r) = \frac{Q_1}{4\pi\varepsilon_0 |r - r_1|}}$$

$V(r)$ is the potential at a distance r from charge Q_1

Superposition of potential fields

For single point charge

$$V(r) = \frac{Q_1}{4\pi\varepsilon_0 |r - r_1|}$$



For system of multiple charges

For Two charges

$$V(r) = \frac{Q_1}{4\pi\varepsilon_0 |r - r_1|} + \frac{Q2}{4\pi\varepsilon_0 |r - r_2|}$$

For general case (where n charges exist)

$$V(r) = \frac{\Sigma Q_1}{4\pi\varepsilon_0 |r - r_1|} + \frac{\Sigma Q2}{4\pi\varepsilon_0 |r - r_2|} + \cdots\cdots \frac{\Sigma Qn}{4\pi\varepsilon_0 |r - r_n|}$$

$$V(r) = \sum_{m=1}^{n} \frac{Q_m}{4\pi\varepsilon_0 |r - r_m|}$$

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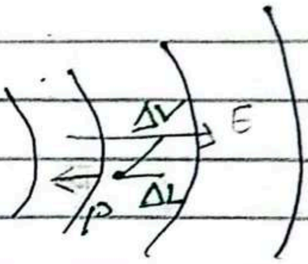
Lecture 19

Topic 4.6 Potential Gradient

General Line Integral relationship

$$V = -\int E \cdot dl$$



$$\Delta V = -E \cdot \Delta L$$
$$= -E \,\Delta L \cos\theta$$

when ΔV is maximized:

→ when $\cos\theta = -1$ $\quad \theta = 180^\circ$ (angle is between E and ΔL)

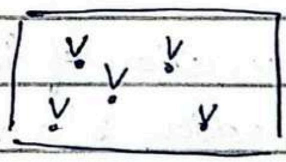
mean ΔL points in the direction opposite to E

(here E is constant dependent on distance)

→ Magnitude of E at any point is given by the maximum rate of change of potential with respect to distance.

Equipotential Surfaces (same potential at every point) (potential difference is zero)

$$\Delta V = E \cdot \Delta L = 0$$
$$= E \cdot \Delta L \cos\theta = 0$$
$$\theta = 90^\circ \qquad \cos 90 = 0$$



→ E cannot be zero

→ ΔL (distance cannot be zero)

→ As ΔV so E also exist

it will be max at $\theta = 0^\circ$ $\quad \cos 0 = 1$

In equipotential surfaces E is always perpendicular to distance (ΔL and E points in direction where potential decreases (if E is outward and if E is inward than potential increases)

$$\Delta V = -E \cdot \Delta L$$

$$E = \frac{\Delta V}{\Delta L} \simeq \frac{dV}{dL}$$

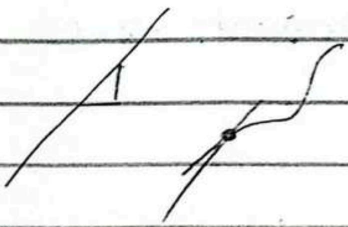
$N \rightarrow$ Normal

$$\therefore E = -\left(\frac{dV}{dL}\right)_{max} a_N = -\frac{dV}{dN} a_N$$

where $\frac{dV}{dL}$ represents the maximum rate of change w.r.t L along the normal direction

N distance

$$E = \left(\frac{dV}{dL}\right)_{max} a_N = -\frac{dV}{dN} a_N$$ where $\frac{dV}{dN}$ is the rate of change of V with respect to L along the normal



Gradient of a Scalar field

The operation of finding E from V is an example of gradient operation which is a vector operation applied to a scalar field

Generic form

$$\text{grad } T = \nabla T = \frac{dT}{dN} a_N$$

from relationship b/w E and V

$$\boxed{E = -\nabla V}$$

In different coordinate system

In RCS

$$\nabla V = \frac{\partial V}{\partial x}\hat{a_x} + \frac{\partial V}{\partial y}\hat{a_y} + \frac{\partial V}{\partial z}\hat{a_z}$$

$$E = -\nabla V = -\left(\frac{\partial V}{\partial x}\hat{a_x} + \frac{\partial V}{\partial y}\hat{a_y} + \frac{\partial V}{\partial z}\hat{a_z}\right)$$

$$E = E_x a_x + E_y a_y + E_z a_z$$

equating both

$$\boxed{E_x = -\frac{\partial V}{\partial x}} \qquad E_y = -\frac{\partial V}{\partial y} \qquad E_z = -\frac{\partial V}{\partial z}$$

In CCS

$$E = -\nabla V = -\left(\frac{\partial V}{\partial \rho} a_\rho + \frac{1}{\rho}\frac{\partial V}{\partial \phi} a_\phi + \frac{\partial V}{\partial z} a_z\right)$$

In SCS

$$E = -\nabla V = -\left(\frac{\partial V}{\partial r} a_r + \frac{1}{r}\frac{\partial V}{\partial \theta} a_\theta + \frac{1}{r\sin\theta}\frac{\partial V}{\partial \phi} a_\phi\right)$$

<u>Given</u>

Q: $V = 2x^2y - 5z$ point $(-4, 3, 6)$

At point P what is V, E, direction of E D an

ρV

$V = 2x^2y - 5z$

$= 2(-4)^2(3) - 5(6)$

$\boxed{V = 66\ V}$

$E = -\nabla V$

$E_x = -\dfrac{\partial V}{\partial x} = -\dfrac{\partial}{\partial x}(2x^2y - 5z)$

$\boxed{E_x = -4xy}$

$E_y = -\dfrac{\partial V}{\partial y} = -\dfrac{\partial}{\partial y}(2x^2y - 5z)$

$\boxed{E_y = -2x^2}$

$E_z = -\dfrac{\partial V}{\partial z} = -\dfrac{\partial}{\partial z}(2x^2y - 5z)$

$\boxed{E_y = 5}$ - E is vector quan

$E = -4xy\,\hat{ax} - 2x^2\,\hat{ay} + 5\hat{ay}$

$= -4(-4)(3)\,\hat{ax} - 2(-4)^2\,\hat{ay} + 5\hat{ay}$.

$E = 48\,\hat{ax} - 32\,\hat{ay} + 5\hat{ay}$

$|E| = \sqrt{48^2 + 32^2 + 5^2} = 57.90\ V/m$

$E = \dfrac{48}{57.90}\,\hat{ax} - \dfrac{32}{57.90}\,\hat{ay} + \dfrac{5}{57.90}\,\hat{ay}$

$\hat{E} = 0.829 a_x - 0.552 a_y + 0.086 \hat{a_z}$

$D = E \varepsilon_0$

$= (48 a_x - 32 a_y + 5 a_z)(8.85 \times 10^{-12})$

$= 4.248 \times 10^{-10} - 2.832 \times 10^{-10} a_y + 4.425 \times 10^{-11} \hat{a_z}$

$\rho_v = \nabla D$

31 oct 2024

Lecture 20

Electric dipole The total potential at point P is sum of potentials due to +Q and −Q

- Distance from +Q to P : R_1
  Distance from −Q to P : R_2

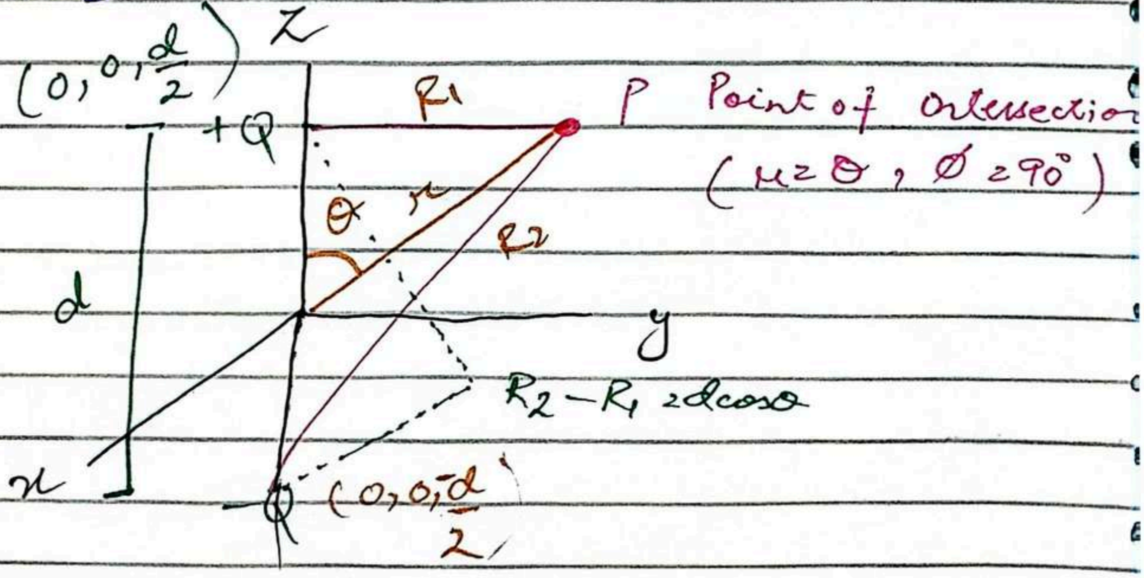
using superposition

$$V = V_{+Q} + V_{-Q}$$

$$= \frac{Q}{4\pi\varepsilon_0 R_1} + \left(\frac{-Q}{4\pi\varepsilon_0 R_2}\right)$$

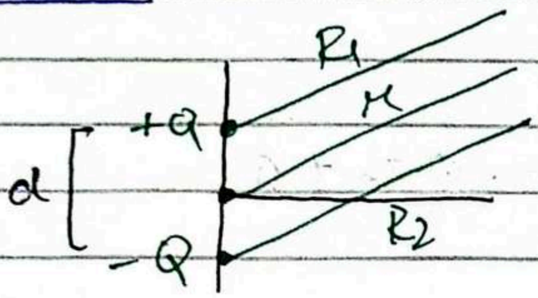
$$= \frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$

$$= \frac{Q}{4\pi\varepsilon_0}\frac{(R_2 - R_1)}{R_1 R_2}$$



New Case

This is in spherical coordinate system



d is a project in spherical coordinate

$$R_1 \approx R_2 \approx r$$

$$V = \frac{Q(R_2 - R_1)}{4\pi\varepsilon_0 R_1 R_2}$$ $r^2 \rightarrow$ origin sa distance

$$\boxed{V = \frac{Q\,d\cos\theta}{4\pi\varepsilon_0 r^2}}$$

$d \rightarrow$ distance b/w two points

$$E = -\nabla V = -\left(\frac{\partial V}{\partial r}\hat{a_r} + \frac{1}{r}\frac{\partial V}{\partial \theta}\hat{a_\theta} + \frac{1}{r\sin\theta}\frac{\partial V}{\partial \phi}\hat{a_\phi}\right)$$

$$E = -\left[\frac{\partial}{\partial r}\left(\frac{Qd\cos\theta}{4\pi\varepsilon_0 r^2}\right) + \frac{1}{r}\left(\frac{\partial}{\partial \theta}\left(\frac{Qd\cos\theta}{4\pi\varepsilon_0 r^2}\right)\right) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}\left(\frac{Qd}{4\pi\varepsilon_0}\right)\right]$$

$$= -\left[-\frac{1}{r} \times \frac{Qd\cos\theta}{4\pi\varepsilon_0} + \frac{1}{r}\left[\frac{Qd}{4\pi\varepsilon_0}\right]\left[-\sin\theta\right] + 0\right)$$

$$= \left[\frac{Qd\cos\theta}{4\pi\varepsilon_0 r} + \frac{Qd\sin\theta}{4\pi\varepsilon_0 r^3}\right]$$

$$E = \frac{Qd}{4\pi\varepsilon_0 r^3}\left(2\cos\theta\,\hat{a_r} + \sin\theta\,\hat{a_\theta}\right)$$

D4.9 D4.10

Dipole moment $\vec{P}$

$$\vec{P} = Q\vec{d}$$

→ where $\vec{d}$ is the vector from $-Q$ to $+Q$

→ units of $\vec{P}$ are C·m

→ $V = \frac{P\cos\theta}{4\pi\varepsilon_0 r^2} \hat{a_r}$

at r we are taking it from origin

$$V = \frac{P\cos\theta}{4\pi\varepsilon_0 r^2} \times \frac{\vec{r}-\vec{r}'}{|r-r'|}$$

other point can be chan if from origin

$r' = 0$

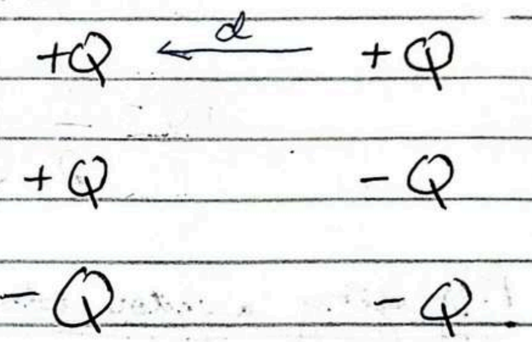
$\theta = 0$

$$V = \frac{\vec{P}}{4\pi\varepsilon_0 |r-r'|^2} \times \frac{\vec{r}-\vec{r}'}{|r-r'|}$$

$V \propto \frac{1}{r^2}$ $\quad E \propto \frac{1}{r^3}$

Energy Density in the Electrostatic field

- Work done by an external source
  ↳ potential energy

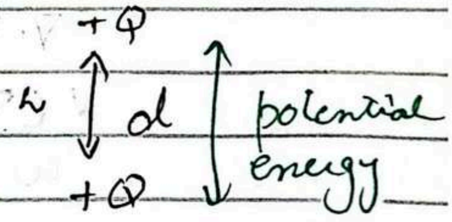


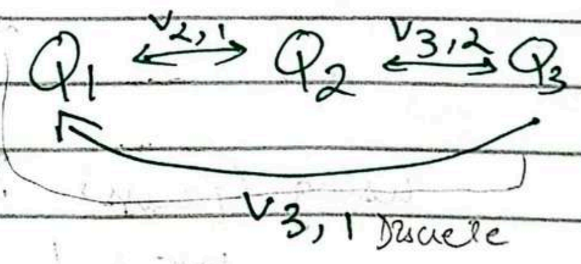
- For doing work on multiple charges

$$W_E = Q_2 V_{2,1} + Q_3 (V_{3,1} + V_{3,2}) + \dots$$

$= C \times \frac{J}{C}$

$= J$





Expression for total PE

$$W_E = \frac{1}{2} \sum_{m=1}^{N} Q_m V_m$$

→ Line charge is continuous

→ ρ_v is continuous

total potential energy W_E of N point charge ρ_{v1} ρ_{v2} discrete

Potential energy for continuous charge distribution

$C/m^3 \quad J/C \quad m^3 = J$

$$W_E = \frac{1}{2} \int_v \rho_v \, V \, dv$$

Deriving Energy Density in terms of E and D

Maxwell first Equation

$$\nabla D = \rho_v$$

$$W = \frac{1}{2}\int_v (\nabla D)\, V\, dv$$

Applying a vector identity

$$\nabla \cdot DV = V(\nabla \cdot D) + D\cdot(\nabla V)$$

$$V(\nabla D) = \nabla\cdot(DV) - D(\nabla V)$$

$$W = \frac{1}{2}\int_v \left[\nabla\cdot(VD) - D\cdot(\nabla V)\right] dv$$

Gauss Law

$$\oint_V \nabla\cdot DV = \oint_S DV\cdot ds$$

using Gauss's Divergence

$$W_E = \frac{1}{2}\underbrace{\oint_{surface} (VD)\cdot ds}_{0 \text{ at infinity}} - \frac{1}{2}\int_{vol} D\cdot(\nabla V)\, dv$$

Assuming the system is isolated and V tend to zero at infinity the surface integral

$$V = \frac{Q}{4\pi r^2}$$

$$V \to 0$$

$$W_E = -\frac{1}{2}\int_{vol} D(\nabla V)\, dv$$

$$E = -\nabla V$$

$$= \frac{1}{2}\int_{vol} D(-\nabla V)\, dv$$

$$\boxed{W_e = \frac{1}{2}\int D\cdot E\, dv}$$

Energy in the electrostatic field

Energy Density

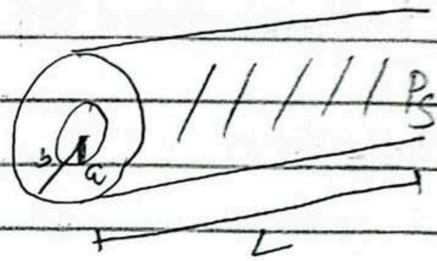
$$\frac{dW_E}{dV} = \frac{1}{2} D \cdot E$$

energy per unit

energy density

Energy Stored in Coaxial Section

$$= W_E = \pi L \frac{a^2 \rho_s^2}{\varepsilon_0} \ln \frac{b}{a}$$



coaxial cable section

Energy stored in capacitor

$$= W_E = \frac{1}{2} QV \qquad Q = CV$$

$$= \frac{1}{2} (CV) V$$

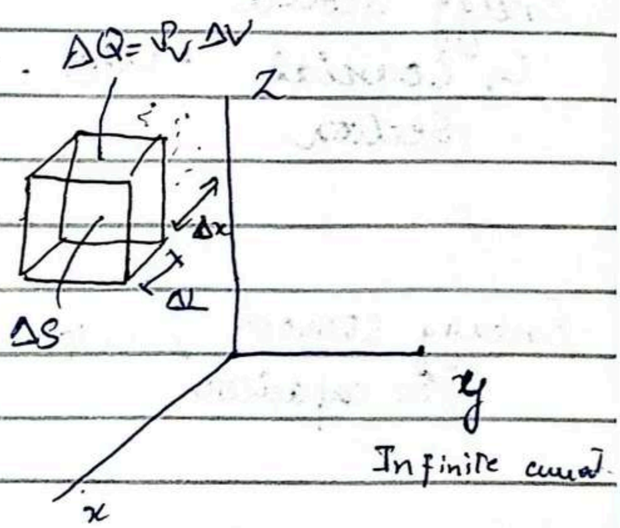
$$= \frac{1}{2} CV^2$$

Lecture 21

Topic 5.1 Current and Current Density

Rate of flow of charges $I = \frac{dQ}{dt}$ Unit Ampere or $\frac{C}{s}$

Direction



Current Density Rate of charge flow through a region (volume)

current density is a 'vector' quantity $\vec{J}$.

$$J = \rho_v V$$

ρ_v → C/m^3, Volume Charge density

V → Velocity of charge carrier, m/s

Unit A/m^2

Relationship with current

The current crossing an infinite surface ds normal to J is

$$dI = J\,ds$$

$$I = \int \vec{J} \cdot ds$$

Current Density and velocity Relationship Consider a volume element of charge

$$\Delta Q = \rho_v \Delta V$$
$$= \rho_v \Delta S \Delta L$$
$$= \rho_v \Delta S \Delta x$$

Assuming the charge moves with velocity 'v' through an incremental distance Δx in time Δt

$$\Delta I = \frac{\Delta Q}{\Delta t} = \frac{\rho_v \Delta S \Delta x}{\Delta t}$$

$$\frac{\Delta x}{\Delta t} = v_x$$

$$\boxed{\Delta I = \rho_v \Delta S v_x}$$

For infinite condition

$$I = \rho_v \cdot v \cdot S \rightarrow \text{surface}$$

$\downarrow$ velocity

$$I = \rho_v \, v \cdot S$$
$$\frac{I}{S} = \rho_v V \qquad J = A/m^2$$

$$\boxed{J = \rho_v \cdot V} \rightarrow \text{convection current density}$$

where volume charge density and velocity comes.

Topic 5.2 Current Continuity

Current continuity is related to charge conservation (charge can neither be created nor be destroyed

Integral and form of continuity Equation

$$I = \int_S J \cdot ds$$

where

J: current density vector (A/m^2)

From the conservation of charge principle, the rate of charge leaving the region must equal the decrease in the charge within the volume of Q_i represents the charge inside the closed surface

$$\oint J \cdot ds = \frac{d Q_i}{dt}$$

The negative sign indicates the outward current corresponds to a decrease in the charge within the enclosed region.

Differential form of continuity Equation

Divergence Theorem $\oint J\, ds = \int_{vol} (\nabla J)\, dv$

$$\int_{VOL} (\nabla J) ds = \frac{-d Q_i}{dt} = \int_S J \cdot ds$$

As Q is charge enclosed $\quad Q = \int P_v dv$

$$\int_V (\nabla J) ds = -\frac{d}{dt} \int P_v dv$$

$$\boxed{\nabla \cdot J = -\frac{\partial}{\partial t} \rho_v}$$

unit: A/m²

Example D5.1

$J = 10\rho^2 z \, a_\rho - 4\rho \cos^2\phi \, a_\phi \; (mA/m^2)$

Find current density at $P(\rho = 3, \phi = 30°, z = 2)$

$J = J_\rho \hat{a_\rho} + J_\phi \hat{a_\phi}$

$J = 10\rho^2 z \hat{a_\rho} - 4\rho \cos^2\phi \, a_\phi$

$= 10(3)^2(2)\hat{a_\rho} - 4(3)\cos^2(30)$

$\boxed{J = 180 \hat{a_\rho} - 9 \, a_\phi \; mA/m^2}$

$\boxed{J_\rho = 180}$ $\boxed{J_\phi = -9}$

Total current

$\rho = 3 \qquad 0 < \phi < 2\pi \qquad 2 < z < 2.8$

$I = \int_s J \, ds$

$ds = \rho \, d\phi \, dz \, a_\rho$

$I = \int_2^{2.8} \int_0^{2\pi} (10\rho^2 z \hat{a_\rho} - 4\rho\cos^2\phi \, a_\phi)(\rho \, d\phi \, dz \hat{a_\rho}) \times 10^{-3}$

$= \int_2^{2.8} \int_0^{2\pi} 10\rho^2 z \, a_\rho (\rho \, d\phi \, dz) a_\rho$

$= \int_2^{2.8} \int_0^{2\pi} 10\rho^3 z \, d\phi \, dz$

$= 10(3)^3 \int_0^{2\pi} d\phi \int_2^{2.8} z \, dz$

$= 270(2\pi - 0)\left[\frac{(2.8)^2 - (2)^2}{2}\right] = 3.867 \; A$

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Lecture 22

Topic 5.3

Metallic Conductor

Material that allows the free Flow of electric charges enabling the conduction of electricity

There are different energy Level within The atom - elections tansition take place b/w | these energy Levels
└ represent sum of KE and PE

Valence band: The band containing the outermost elections

Conduction band: Higher energy levels where free election moment occur

Conductor Close to each other

Semi conductor Vacant spaces

Insulator Large difference elections cannot freely move

Conductors No gap b/w valence and conduction band elections moves freely

Semi conductor A small energy gaps exists b/w valence and conduction bands

Insulator Large energy gap b/w valence and conduction band

Conduction in Metallic Conductors

Election motion in a field

$$F = qE$$

$$\because q = -e$$

$$\boxed{F = -eE}$$

Drift Velocity

$$v_d = -\mu_e E$$

μ_e : Mobility of elections (measured in m^2/V)

Current Density

$$J = -\rho_e \mu_e E$$

ρ_e : Free election density

J : Current Density (A/m^2)

Ohm's Law (in Terms of current density)

$$J = \sigma E$$

$$\sigma = -\rho_e \mu_e$$

σ conductivity (S/m, siemens per meter)

$$V = IR$$

For uniform field

$$R = \frac{L}{\sigma S}$$ $S \rightarrow$ surface area

$$V = \frac{L}{(\sigma S)} I$$

conductivity

$$V = \left(\frac{L}{\rho e \mu e S}\right) I$$

for non uniform field

$$R = \frac{-\int_a^b E\,dl}{\int_S \sigma E\,ds}$$

<u>Effect of temperature</u>

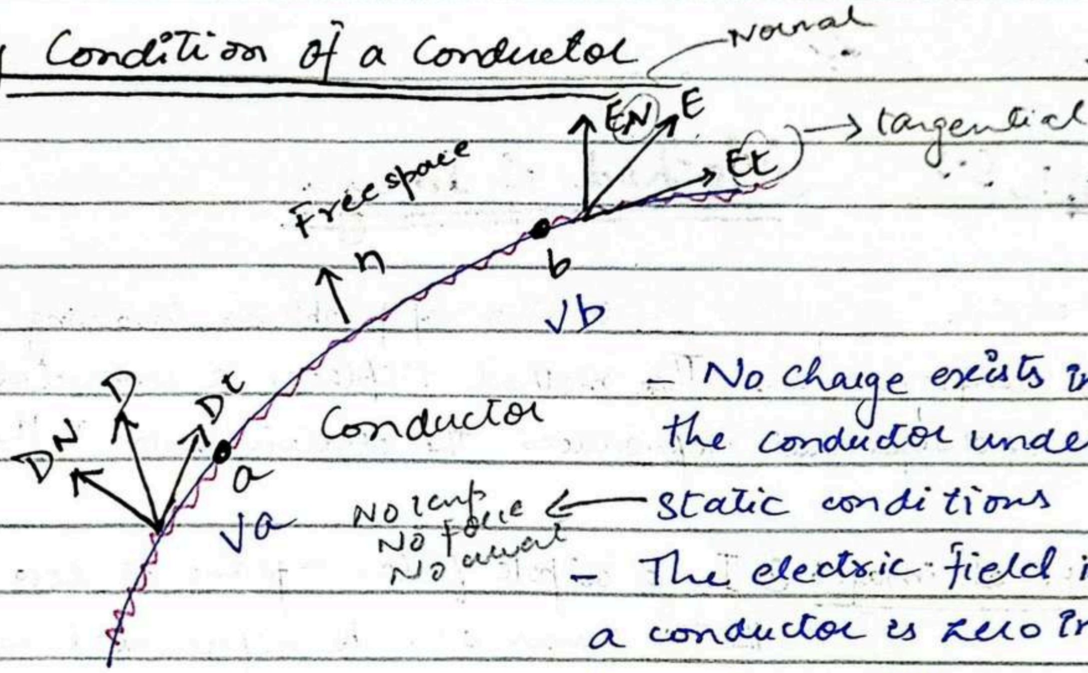
- Increases lattice vibration
- Decrease mobility
- Decrease conductivity (σ)
- Increase R (inverse relation b/w conductivity & R)

$\uparrow T$

$$\rho = \frac{1}{\sigma}$$

$$\text{Resistivity} = \frac{1}{\text{conductivity}}$$

Boundary Condition of a Conductor



charges distributed evenly inside conductor
On surface it is no evenly distributed

- No charge exists inside the conductor under static conditions
- The electric field inside a conductor is zero in static condition

(1) $E_t = 0$ → Tangential electric field

If $E_t \neq 0$ → violating the static condition

(2) $E_N \varepsilon_0 = \rho_s$ (surface charge density) → Line Charge ρ_L

$$\boxed{E_N = \frac{\rho_s}{\varepsilon_0}}$$ → permittivity of free space

(3) Electric flux Density

$D = E\varepsilon_0$

$D_N = E_N \times \varepsilon_0$

$D_N = \frac{\cancel{\varepsilon_0}}{\cancel{\varepsilon_0}} \times \rho_s$

$D_N = \rho_s$

$D_t = 0$

$\boxed{D_N = \rho_s}$

(4) Equipotential Surface

2 points potential difference is equal to zero

$V_a = V_b$

$D_t = E_t \times \varepsilon_0$

$\boxed{D_t = 0}$ as $E_t = 0$

Electric field is also related to potential

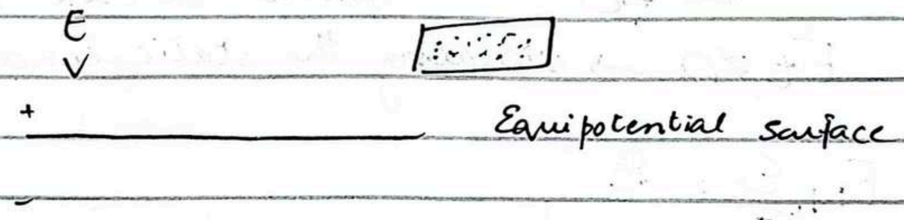
14 Nov 2024

Lecture 22

Topic 5.5 Method of Images

The technique simplifies the solution of problems involving conductors and charge configurations. This method replaces a conductor by an equivalent system of charges that reproduces the electric field in the region of interest.

Concept of images ① In a dipole field, a plane at zero potential b/w the two charges can be represented as an infinite conducting plane



② Instead of analyzing the system with the conducting plane we can replace it with a mirror image of charges on the other side of the plane

③ The image charge is opposite in magnitude and located symmetrically with conducting plane

point charge $Q \longrightarrow$ its image charge $-Q$

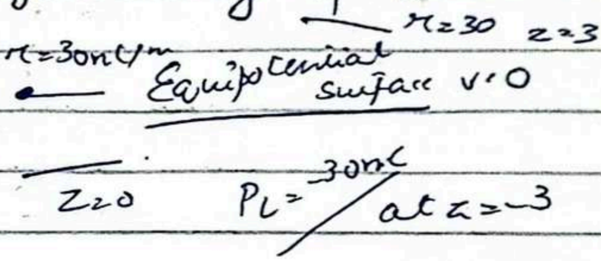
$$V = \frac{Q}{4\pi\varepsilon_0|r-r_1|} + \frac{-Q}{4\pi\varepsilon_0|r-r_2|}$$

$$= \frac{Q}{4\pi\varepsilon_0|r-r_1|} - \frac{Q}{4\pi\varepsilon_0|r-r_2|}$$

D5.6

$\rho_L = 30n\ C/m$ at $z=3m$ along the $x=0$ axis. The conducting plane lies at $z=0$. Find ρ_S at $P(2,5,0)$.

Sol: Replace the conducting plane with an image line charge of $\rho_L = -30n\ C/m$ at $z=-3$



From the +ve line charge to $P(2,5,0)$: $R_+ = 2\hat{a}_x - 3\hat{a}_z$

From the -ve line charge to $P(2,5,0)$: $R_- = 2a_x + 3a_z$

$$E_+ = \frac{\rho_L}{2\pi\varepsilon_0 R_+^2} \times \frac{\vec{R}_+}{|R_+|} \qquad E_- = \frac{-\rho_L}{2\pi\varepsilon_0 R_-^2} \frac{\vec{R}_-}{|R_-|}$$

$$E = E_+ + E_- = -249\,a_z\ V/m$$

$$D = E\varepsilon_0 = -2.20\,a_z\ nC/m^2 = \rho_S$$

$\rho_S = D$ if it is directed toward the plane

Electro Magnetic Theory : Date: 21 NOV 2024

Lecture 24

Chapter 6 Capacitance

Capacitance is a measure of a systems ability to store electric charge

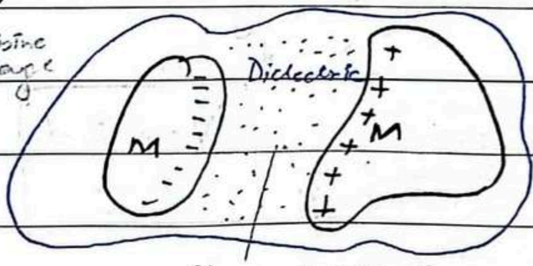
It is defined in terms of two oppositely charged conductors separated by a dielectric medium

→ Electric field b/w conductor is normal to the surface

→ Each conductor is an equipotential surface The potential difference V_0 exists b/w two conductor

Capacitance 'C' is define as the ratio of the total charge on either conductor to the potential difference b/w

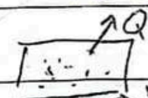
$$\boxed{C = \frac{Q}{V_0}}$$ → Combine charge



Charge is stored b/w plate & dielectric medium

Q : Determined by the surface integral over the positive conductor

V_0 : Calculated by integrating the electric field along a path from the negative to the positive



due to surface charge

$$C = \frac{\oint_S E \cdot dS}{-\int_{-}^{+} E \cdot dl}$$

→ Charge (in terms of electric field)

→ potential (in term of electric field)

along the surface potential

unit SI : unit of capacitance Farad (F)

Topic 6.2 Parallel Plate Capacitor

Consisting of two parallel conducting plates separated by a uniform dielectric medium

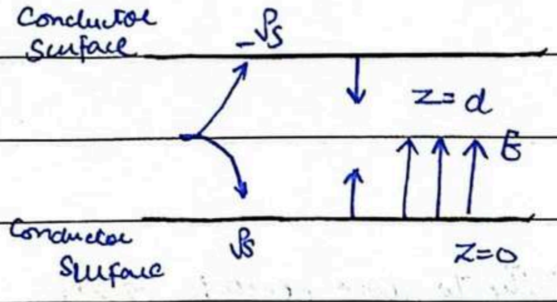


Plate separation : d

Plate area : S

Dielectric permittivity : ϵ

Electric flux density $D = \rho_s \hat{az}$ where ρ_s is the charge density

$$E = \frac{D}{\varepsilon} = \frac{\rho_s \hat{az}}{\varepsilon}$$

$$V_0 = -\int_d^0 E \cdot dl = -\int_d^0 \frac{\rho_s}{\epsilon} \cdot dl$$

reference point always w.r.t to it ← $\boxed{V_0 = \frac{\rho_s d}{\varepsilon}}$ -ve sign is dependent on reference point

if positive to negative than -ve

if negative to positive than +ve

$$\boxed{V_0 = \frac{\rho_s d}{\varepsilon}}$$

↳ +ve → $+\rho_s$ → $z=0$

$$C = \frac{\epsilon S}{d}$$

$$\boxed{V_0 = \frac{-\rho_s d}{\varepsilon}}$$

↳ -ve → $-\rho_s$ → $z=d$

$$C = \frac{Q}{V}$$

$$Q = C/m^2 \times m^2 = \rho_s \times S$$

$$C = \frac{Q}{V} = \frac{\rho_s S}{\frac{\rho_s d}{\varepsilon}} = \frac{\varepsilon S}{d}$$

Date: ______

$$\boxed{C = \frac{\epsilon S}{d}}$$

→ it doesnot depend on surface charge density

→ it depend on ε, S and d

→ d ↑ capacitance ↓

Problem: $\varepsilon_r = 6$ $A = 10\,m^2$ $d = 0.01\,m$

$\varepsilon = \varepsilon_0 \varepsilon_r$

$$C = \frac{\varepsilon S}{d} = \frac{\varepsilon_0 \varepsilon_r (S)}{d} = \frac{8.85 \times 10^{-12} \times 6 \times 10}{0.01}$$

$$\boxed{C = 5.31 \times 10^{-8} F}$$

Energy stored in a capacitor

$$\boxed{W_E = \frac{1}{2} C V_0^2 = \frac{1}{2} Q V_0 = \frac{1}{2} \frac{Q^2}{C}}$$

↳ this is potential ↳ stored

$$\boxed{C = \frac{\varepsilon S}{d}}$$

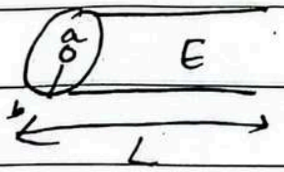
if ε ↑ C ↑ W_E ↑

if A ↑ C ↑ W_E ↑ ↑C ε↑

if V ↑ C ↑ W_E ↑

Capacitance **Use case 1**

– Coaxial cable



$$E_r = \frac{Q/L}{2\pi \varepsilon r} = \frac{\lambda}{2\pi \varepsilon r}$$

$$\int \frac{dr}{r} = \ln\left(\frac{b}{a}\right)$$

$$V_0 = \int_a^b E_r\,dr = \frac{\lambda}{2\pi\varepsilon} \ln\left(\frac{b}{a}\right)$$

CamScanner

Date:

$$C = \frac{Q}{V_0} = \frac{2\pi \varepsilon L}{\ln\left(\frac{b}{a}\right)}$$

$$V_0 = \frac{\lambda}{2\pi\varepsilon} \ln\left(\frac{b}{a}\right)$$

$$V_0 = \frac{Q/L}{2\pi\varepsilon} \ln\left(\frac{b}{a}\right)$$

$$\frac{Q}{L} = \frac{V_0 \times 2\pi\varepsilon}{\ln(b/a)}$$

$$\frac{Q}{V_0} = \frac{L\, 2\pi\varepsilon}{\ln(b/a)}$$

$$\boxed{C = \frac{Q}{V_0} = \frac{L \times 2\pi\varepsilon}{\ln(b/a)}}$$

Date: 28-Nov 2024

Lecture:

Electromagnetic Theory

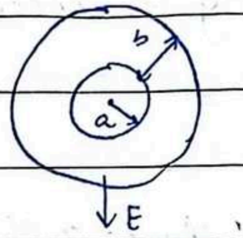
Topic 6.3

Spherical Capacitor

Inner radius 'a'

Outer radius 'b'

Permitivity 'ε'



$$E_r = \frac{Q}{4\pi \varepsilon r^2}$$

$$V_0 = \int_a^b E_r dr$$

$$V_0 = \frac{Q}{4\pi \varepsilon_0} \int_a^b \frac{1}{r^2} dr$$

$$= \frac{-Q}{4\pi \varepsilon_0} \left[\frac{1}{b} - \frac{1}{a} \right]$$

$$\boxed{V_0 = \frac{Q}{4\pi \varepsilon} \left[\frac{1}{a} - \frac{1}{b} \right]}$$

$$\boxed{C = \frac{Q}{V_0} = \frac{4\pi \varepsilon ab}{b-a}}$$

Special case $b \longrightarrow \infty$

$$\boxed{C = 4\pi \varepsilon a}$$

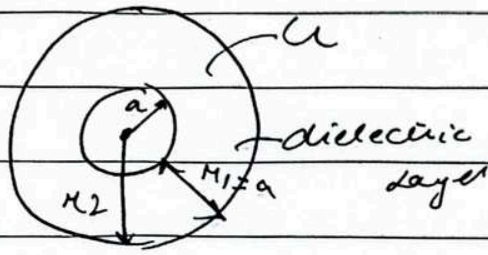
Date:

Special Capacitor with dielectric layer

Sphere with radius → a

dielectric layer thickness

$= r_1 - a$



Outer conductor radius r_2

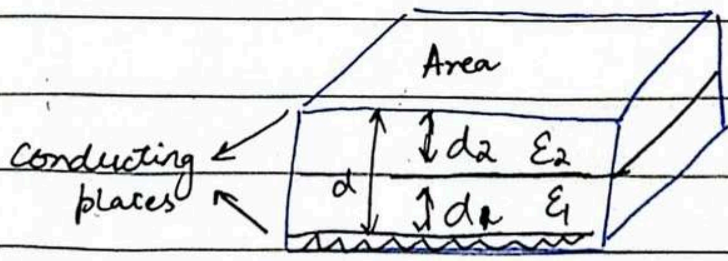
$$C = \frac{4\pi}{\frac{1}{\varepsilon_r}\left(\frac{1}{a} - \frac{1}{r_1}\right) + \frac{1}{\varepsilon_0 r_1}}$$

$$E_r = \frac{Q}{4\pi\varepsilon_0 r^2} \qquad (a < r < r_1)$$

$$E_r = \frac{Q}{4\pi\varepsilon_0 r^2} \qquad (r_1 < r)$$

$$V_a - V_\infty = \frac{Q}{4\pi}\left[\frac{1}{\varepsilon_r}\left(\frac{1}{a} - \frac{1}{r_1}\right) + \frac{1}{\varepsilon_0 r_1}\right]$$

Parallel Plate Capacitor with multiple dielectric



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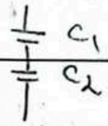
$C = \frac{\varepsilon S}{d}$

Capacitor are in series i.e.



$$d = d_1 + d_2$$

$$C_1 = \frac{\varepsilon_1 S}{d_1}$$

$$C_2 = \frac{\varepsilon_2 S}{d_2}$$

$$C_{Total} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$= \frac{C_1 C_2}{C_1 + C_2}$$

$$= \frac{\left(\frac{\varepsilon_1 S}{d_1}\right)\left(\frac{\varepsilon_2 S}{d_2}\right)}{\frac{\varepsilon_1 S}{d_1} + \frac{\varepsilon_2 S}{d_2}} = \frac{\left(\frac{\varepsilon_1 \varepsilon_2}{d_1 d_2}\right) S^2}{\frac{\varepsilon_1 d_1 S + \varepsilon_2 d_2 S}{d_1 + d_2}} = \frac{\frac{(\varepsilon_1 \varepsilon_2) S}{d_1 d_2}}{\varepsilon_1 d_1 + \text{[illegible]}}$$

$$= \frac{\frac{d_1 + d_2}{\varepsilon_1 d_1 S + \varepsilon_2 d_2 S}}{\left(\frac{\varepsilon_1 \varepsilon_2}{d_1 d_2}\right) S^2}$$

$$C_{Total} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2}} = \frac{1}{\frac{1}{\frac{\varepsilon_1 S}{d_1}} + \frac{1}{\frac{\varepsilon_2 S}{d_2}}}$$

$$= \frac{1}{\frac{d_1}{\varepsilon_1 S} + \frac{d_2}{\varepsilon_2 S}}$$

$$= \frac{1}{\frac{1}{S}\left(\frac{d_1}{\varepsilon_1} + \frac{d_2}{\varepsilon_2}\right)} = \frac{S}{\frac{d_1}{\varepsilon_1} + \frac{d_2}{\varepsilon_2}}$$

Date:

$$= \frac{S}{\frac{d_1\varepsilon_2 + d_2\varepsilon_1}{\varepsilon_1\varepsilon_2}}$$

$$\boxed{C_{total} = \frac{S\,\varepsilon_1\varepsilon_2}{d_1\varepsilon_2 + d_2\varepsilon_1}}$$

1 in = 0.3048 m

Q. Coaxial Cable capacitor

1 ft = 12 in = 12 × 0.308 m

Inner conductor diameter = 0.1045 in = 0.0318

Outer = 0.680 in = 0.2072

Dielectric $\varepsilon_r = 2.26$ Length $L = 1$ ft = 3.696

C = ?

$$C = \frac{2\pi\varepsilon_0\varepsilon_r L}{\ln\left(\frac{b}{a}\right)}$$

$$= \frac{2\pi(8.85\times10^{-12})(2.26)(3.696)}{\ln\left(\frac{0.2072}{0.0318}\right)}$$

$$= 2.475\times10^{-10}$$

$$\boxed{C = 0.2475\,nF}$$

Parallel plate capacitor

Area = 1 cm × 4 cm = 4 cm^2

$\varepsilon_1 = 1.5$ $\varepsilon_2 = 2.5$ $\varepsilon_3 = 6$

$d_1 = d_2 = d_3 = 0.1$ mm

$$C_{Total} = \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right)^{-1} = \left(\frac{d_1}{\varepsilon_1 S} + \frac{d_2}{\varepsilon_2 S} + \frac{d_3}{\varepsilon_3 S}\right)^{-1}$$

$$= \frac{1}{\frac{d_1}{\varepsilon_0 \varepsilon_1 S} + \frac{d_2}{\varepsilon_0 \varepsilon_2 S} + \frac{d_3}{\varepsilon_0 \varepsilon_3 S}}$$

$$= \frac{1}{\frac{0.1\times10^{-3}}{8.85\times10^{-12}\times1.5\times4\times10^{-4}} + \frac{0.1\times10^{-3}}{8.85\times10^{-12}\times2.5\times4\times10^{-4}} + \frac{0.1\times10^{-3}}{8.85\times10^{-12}\times6\times4\times10^{-4}}}$$

$$\boxed{C = 24nF}$$

Electromagnetic Theory

Date: 28 Nov

Lecture: Laplace Equation / Poisson Equation

Gauss Law

$$\nabla D = \rho_v$$

$$D = \varepsilon E$$

$$E = -\nabla V$$

$$\boxed{D = \varepsilon(-\nabla V)}$$

$$\nabla(-\nabla V)\varepsilon = \rho_v$$

$$\boxed{\nabla \cdot \nabla V = -\frac{\rho_v}{\varepsilon}} \rightarrow \text{Poission Equation}$$

$$\nabla V = \frac{\partial V}{\partial x}\hat{a_x} + \frac{\partial V}{\partial y}\hat{a_y} + \frac{\partial V}{\partial z}\hat{a_z}$$

$$\nabla \cdot \nabla V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial x^2} = -\frac{\rho_v}{\varepsilon}$$

If $\rho_v = 0$ → zero volume charge density but there is a possibility
↓
its a point charge or line charge

Then

$$\nabla \cdot \nabla V = \boxed{\nabla^2 V = 0} \quad \text{Laplace equation}$$

∇^2 is called Laplace of V

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

Date:

Q, Calculate V and ρ_v at point 'P'

a) $V = \frac{4yz}{x^2+1}$ at $P(1,2,3)$

$= \frac{4(2)(3)}{1^2+1}$

$V = 12$

b) $V = 5\rho^2 \cos 2\phi$ $(\rho = 3, \phi = \frac{\pi}{3})$

$V = 5(3)^2 \cos\left(2 \times \frac{\pi}{3}\right)$

$V = 44.96\,V$ (in degree) $= -22.5$ (radian)

(part a)

$\rho_v = ?$

$\nabla \nabla V = \frac{-\rho_v}{\varepsilon}$

$\nabla V = \frac{\partial}{\partial x}\left(\frac{4yz}{x^2+1}\right) + \frac{\partial}{\partial y}\left(\frac{4yz}{x^2+1}\right) + \frac{\partial}{\partial z}\left(\frac{4yz}{x^2+1}\right)$

$= \frac{\partial}{\partial x}\left(4yz\,(x^2+1)^{-1}\right) + \frac{4z}{x^2+1} + \frac{4y}{x^2+1}$

$= (-1)\,4yz\,(x^2+1)^{-2}(2x) + \frac{4z}{x^2+1} + \frac{4y}{x^2+1}$

$= -\frac{8xyz}{(x^2+1)^2} + \frac{4z}{x^2+1} + \frac{4y}{x^2+1}$

$= -\frac{8xyz}{(x^2+1)^2} + \frac{4y + 4y}{x^2+1}$

Date:

$$\nabla^2 V = -\frac{16yz}{(x^2+1)^3}(3x^2+1)$$

$$\nabla^2 V = -48$$

$$\rho_v = -\varepsilon \nabla^2 V$$

$$= -106.2 \ pC/m^2$$

Steps

1/ Given V use $E = -\nabla V$ find E

2/ use $D = \epsilon E$ to find D

3/ Evaluate D at either capacitor plate $D = D_s = D_n a_n$

4/ $\rho_s = D_n$

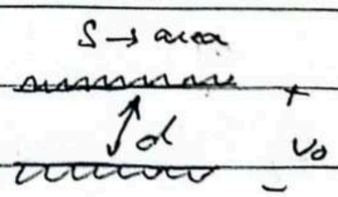
5/ Find Q by surface integration over the capacitance

$$Q = \int_s \rho_s \, ds$$

6/ $C = \dfrac{Q}{V}$

Parallel plate capacitor of area S separation d, and potential difference V_0 b/w plates

Laplace equation $\dfrac{d^2V}{dx^2} = 0$



$$\frac{dV}{dx} = A$$

$V = Ax + B \Rightarrow V = \dfrac{V_0 x}{d}$ $\quad B = 0$ for parallel plate

$\boxed{A = V_0/d} \Rightarrow$ Slope

1. $E = -\nabla V = -\frac{V_0}{d} a_x$

2. $D = -\frac{\varepsilon V_0}{d} a_x$

3. $D_S = D|_{x=0}$ (only x component)

4. $\rho_s = D_S = -\frac{\varepsilon V_0}{d} a_x$

5. $Q = \int_S -\frac{\varepsilon V_0}{d} dS$

$= -\frac{\varepsilon V_0}{d} \int_S dS$

$Q = -\frac{\varepsilon V_0 S}{d}$

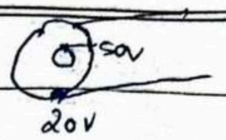
6. $C = \frac{|Q|}{V_0} = \frac{\frac{\varepsilon_0 S V_0}{d}}{V_0} = \frac{\varepsilon S}{d}$

D 6.6 Find $|E|$ at $P(3,1,2)$ in RCS for the field of

a) two coaxial conducting cylinders $V = 50V$ at $\rho = 2m$ and $V = 20V$ at $\rho = 3m$

b) two radial conducting planes $V = 50V$ at $\phi = 10°$ and $V = 20V$ at $\phi = 30°$

Date:

a) For coaxial cylinders 

$$V(\rho) = -\frac{A}{\ln(b/a)} \ln(\rho) + C$$

slope (A), intercept (C), → given logi

$a = 2m \quad b = 3m \quad V(a) = 50V \quad V(b) = 20V$

ρ is varying between inner and outer radius

$$50 = -\frac{A}{\ln(3/2)} \ln(2) + C \rightarrow (1)$$

$$20 = -\frac{A}{\ln(3/2)} \ln(3) + C \rightarrow (2)$$

$$50 = -1.709A + C$$

$$20 = 2.7095A + C$$

$$\boxed{C = 20 - 2.7095A}$$

$$50 = -1.709A + 20 - 2.7095$$

$$50 = -4.418A + 20$$

$$30 = -4.418A$$

$$A = 30 \quad C = 40$$

$$V(\rho) = -30 \ln \rho + 40$$

$$E = -\nabla V = -\frac{\partial V}{\partial \rho} = -\frac{\partial}{\partial \rho}(-30 \ln \rho + 40)$$

$$E = \frac{30}{\rho} \qquad \rho = \sqrt{3^2 + 1^2} = \sqrt{10}$$

$$= \frac{30}{\sqrt{10}} = 9.48$$